# The Generalised Collatz Multiverse

**Symmetries, reductions and fates of the maps $n \mapsto n/2$ (even), $n \mapsto an + b$ (odd)**

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**Abstract**

For integers $a, b \geq 0$ let $T_{a,b}$ send $n$ to $n/2$ when $n$ is even and to $an + b$ when $n$ is odd, so that $T_{3,1}$ is the Collatz map. We give a structure theory for this two-parameter family, over $\mathbb{Z}$.

Two symmetries and one dichotomy account for all of it. The pair $(2a, 2b)$ is a time change of $(a, b)$; the map $n \mapsto dn$ embeds $(a, b)$ into $(a, db)$ for odd $d$; and for each odd prime $p$ the quantity $\min(v_p(n), v_p(b))$ either climbs to $v_p(b)$ and sticks, when $p \mid a$, or is exactly conserved, when $p \nmid a$. We show these are the *only* affine equivalences, so the multiplier $a$ can never change; that they form a terminating, confluent rewriting system whose normal form is always coprime; and that after reduction every universe is one of four families we solve completely, or else has $a \geq 3$ odd, $b$ odd and $\gcd(a, b) = 1$, where the problem is open. Exactly $1/3$ of all universes are open, and $2/\pi^2$ are open and irreducible; the rest are either settled or are a smaller universe in disguise.

The conserved valuations partition a coprime universe into $\tau(b)$ mutually unreachable strata indexed by the divisors of $b$, which decomposes its cycle set exactly. We also determine when 1 itself is periodic, and sharpen the row $a = 1$ enough to make its cycle census exact rather than a lower bound. Two further chapters ask what survives modulo $q$ rather than modulo 2, and what survives when the branch is chosen by a property that is not a congruence at all; the answer to the second is essentially nothing, and we argue that this, rather than any difficulty of degree, is what separates $3x + 1$ from its neighbours.

The framing and many of the phenomena are due to carykh and to Artem, who found them experimentally; we attribute each result and correct three conjectures, with counterexamples in $2x + 1$, $5x + 1$ and $3x + 7$. Every theorem is accompanied by a machine-run check, over $3 \times 10^6$ cases in total.

## 1 Introduction

The Collatz conjecture concerns the map that halves even numbers and sends an odd $n$ to $3n+1$. Its resistance to proof is famous, and Erdős' assessment — that mathematics is not ready for such problems — is still quoted. Less often asked is what happens when the constants 3 and 1 are changed. Writing

$$T_{a,b}(n) \;=\; \begin{cases} n/2 & n \text{ even}, \\ an + b & n \text{ odd}, \end{cases} \qquad a \in \mathbb{Z}_{\geq 0},\; b \in \mathbb{Z},$$

one obtains a two-parameter family — a *multiverse*, in the coinage of the YouTube mathematician carykh [1], whose video animates the $6 \times 6$ grid $0 \leq a, b \leq 5$ and observes that most of its cells

*This paper, the accompanying software and the verification suite were produced in collaboration with an artificial-intelligence system; see the note on the use of artificial intelligence in Section 17.

are in some way degenerate. A follow-up ramble [2] and a reply video by Artem [3] push the observations further, in particular into the row $a = 3$.

Those three sources are experimental. They exhibit striking regularities — that even-indexed cells duplicate their halves, that $3x + 3$ is a scaled copy of $3x + 1$ but $3x + 5$ is not, that $3x + 9$ takes two odd steps to become a copy where $3x + 3$ takes one, that composite offsets contain a copy of every divisor — but they state them as patterns observed rather than as theorems, and they leave the boundary between what is solved and what is open unmapped. This paper draws that boundary.

**What is proved here.**

1. **Rigidity** (Theorem 3.5). The only affine maps intertwining two members of the family are $n \mapsto dn$ with $d$ an odd integer, and they satisfy $a' = a$, $b' = db$. So the multiplier $a$ is a complete invariant of a row: no affine change of variable will ever relate $3x + 1$ to $5x + 1$. In particular the relation between $(2a, 2b)$ and $(a, b)$ — carykh's "even parallel universes" — is *not* a conjugacy; it is a time change, and we give its exact form (Proposition 3.1).

2. **$p$-adic stratification** (Theorem 4.1). For an odd prime $p$, the behaviour of $v_p$ along an orbit is governed entirely by whether $p$ divides $a$. If it does, $\min(v_p(n), v_p(b))$ strictly increases at every odd step until it reaches $v_p(b)$, so $p^{v_p(b)}\mathbb{Z}$ is a global attractor; if it does not, $\min(v_p(n), v_p(b))$ is exactly conserved, so the state space splits. This is the theorem behind Artem's connected/disconnected dichotomy and behind carykh's "factor loyalty".

3. **Normal form** (Theorem 5.2). Halving and attraction form a terminating, confluent rewriting system; the normal form $\mathrm{nf}(a, b)$ is unique and always satisfies $\gcd = 1$. Both rules preserve the fate of every orbit and act bijectively on cycles.

4. **Master classification** (Theorem 6.10). Every universe is exactly one of: constant ($a = 0$), pure ($b = 0$), parity runaway ($a + b$ odd), descent ($a = 1$), or primitive. The first four are solved here, completely and with explicit cycle data. The fifth is open.

5. **Densities** (Theorem 7.4). The open universes have density exactly $1/3$; the open *irreducible* ones have density $2/\pi^2$.

6. **Stratification and cycle decomposition** (Theorem 8.2, Corollary 8.3). A coprime universe partitions into $\tau(b)$ invariant strata indexed by the divisors of $b$, and its cycle set is the corresponding disjoint union $\bigsqcup_{d|b} d \cdot \mathrm{Cyc}^*(T_{a,b/d})$.

7. **Over $\mathbb{Z}$** (Proposition 2.2). The positive integers are invariant exactly when $a + b \geq 1$ and the negative ones exactly when $b \leq a - 1$; sign changes only at odd $n$ with $|n| \leq |b|/a$, so every orbit changes sign at most once. The negative half of $3x + 1$ is the positive half of $3x - 1$.

8. **The cycle through 1** (Theorem 6.7). It is periodic exactly when its forward orbit meets a power of two; $a + b = 2^k$ is the one-step case, and accounts for 93 of the 147 instances we found.

9. **Two chapters outward** (Sections 12 and 13). Modulo $q$: both symmetries survive, the parity obstruction becomes a residue graph, the threshold $a < 4$ becomes $\prod a_i < q^q$, and rigidity fails. Beyond congruences: the theory applies to eventually periodic branch predicates and to nothing else, which we argue is the real reason $3x + 1$ is tractable at all.

**Software.** Everything computational here is produced by the `Multiverse` package. Its modules are `multiverse.py` for the two-parameter family, `qary.py` for Sections 12 and 13, and `literature.py` for Appendix A; `generate_data.py` drives them, and `Collatz_Program.py` is the interactive front end. All of it lives in the repository [43]. Section 10 describes them and Section 11 reports the verification suite. No number in this paper was transcribed by hand.

### 1.1 Related work

The $3x+1$ problem itself has an enormous literature, surveyed by Lagarias [38, 39] and catalogued in his annotated bibliographies [15, 16, 17]. We use only a few landmarks: Terras [40] showed almost every $n$ has finite stopping time; Eliahou [11] constrained the possible periods of a nontrivial cycle; Tao [41] showed almost all orbits attain almost bounded values; Hercher [12] ruled out $m$-cycles for $m \leq 91$; and Barina [13] pushed numerical verification past $2^{71}$. Gonçalves, Greenfeld and Madrid [14] extended Tao's method to a class of generalized maps.

Generalisation of the constants is older than it looks. Crandall [4] introduced the $qx+1$ maps in 1978 and devoted a section to the general $qx+r$; his conjecture — that for odd $q > 3$ the map $qx+1$ has an orbit not reaching 1 — was proved for almost all $q$ in the sense of asymptotic density by Franco and Pomerance [5]. Santos [6] treats $qn+1$ extensively, establishing finiteness of the periodic orbits and settling $q = 2^k - 1$. Belaga and Mignotte [7] study $3x+d$, which is the row $a = 3$ of our grid. Matthews [8] develops a different generalisation, with several branches and a Markov-chain/ergodic analysis, closer to the $q$-ary map of Section 2.1 than to the two-parameter family here.

Two facts from that literature bound what this paper can hope to do. First, Conway [9] showed that generalized Collatz functions are universal, and Kurtz and Simon [10] that the associated reachability problem is $\Pi^0_2$-complete; so no classification theorem can cover the unrestricted $q$-ary family. Second, Franco and Pomerance proved Crandall's conjecture — that $qx+1$ admits an orbit never reaching 1 — for almost all odd $q$. So along the column $b = 1$ the rows $a \geq 5$ are not merely unproven: for almost all such $a$ the Collatz-style statement is known to be *false*. Our contribution is orthogonal to both: it is a structure theory for the two-parameter slice, saying which of its members are the same problem and which are genuinely distinct.

Finally, the multiverse framing that prompted this paper is due to carykh [1, 2] and Artem [3], and the community site collatzresearch.org [18] collects related experimental work. Section 16 attributes each individual result.

### 1.2 Prior work on the generalised maps

Because this paper claims a structure theory for a family that has been studied for fifty years, we owe the reader a precise account of what was already known. We surveyed the corpus catalogued by ccchallenge.org [19] — 375 papers at the time of writing — and classified it (Appendix A); 49 of those papers concern generalised maps. The following are the ones that overlap this paper.

**The column $b = 1$.** Crandall [4] posed the $qx+1$ problem; Steiner [20, 21] studied $Qx+1$ for odd $Q$ and Franco [22] connected the cycle problem to Diophantine approximation, which is the analytic content behind our Theorem 8.1: a cycle needs $2^m/a^k$ close to 1, hence needs a good rational approximation to $\log_2 a$. Franco and Pomerance [5] then proved Crandall's conjecture for almost all $q$ in the sense of asymptotic density, and Santos [6] established finiteness of the cycle set for each $q$ and settled $q = 2^k - 1$. Our row $b = 1$ (Table 7) is a census inside their setting, not a competitor to it.

**The row $a = 3$.** This is the closest prior work, and it anticipates part of Section 5. Belaga and Mignotte [7] embed $3x+1$ in the $3x+d$ family and — crucially — restrict throughout to $d$ coprime to 6, that is, $d$ odd and not divisible by 3. That restriction is exactly our normal form specialised to $a = 3$: by Theorem 5.2 the even $d$ reduce by halving and the multiples of 3 reduce by attraction, so nothing is lost. They state the $3x+d$ *conjecture* — that each such $d$ admits only finitely many cycles — and their Proposition 3.1, that the fixed points of the accelerated map are $n = d/(2^k - 3)$, is the $k = 1$ case of the cycle equation. Belaga [23] gives polynomial upper bounds on the number and size of $(3x + d)$-cycles of given odd length, and Belaga and Mignotte [24] tabulate the *primitive* cycles of $3x + d$ — their term, and their notion agrees with ours: a cycle no element of which shares a factor with $d$. Simons [25] bounds $m$-cycles for generalised Syracuse sequences.

We have not found the divisor decomposition (Corollary 8.3) stated as a general theorem anywhere, though it is visible in the examples of [7]: their nontrivial fixed points of $S_{65}$ are 5 and 13, which in our language are 5 times the fixed point of $S_{13}$ and 13 times that of $S_5$. What we add is that these copies *exhaust* the cycle set, and that the same decomposition holds for the whole phase space (Theorem 8.2).

**The $q$-ary maps.** Section 12 is the elementary layer of a subject with a substantial theory. Matthews and Watts [26, 27] generalised Hasse's version of the Syracuse algorithm to exactly the shape of Section 2.1 and analysed it with Markov chains; Leigh [28] and Buttsworth and Matthews [29] developed the Markov-matrix machinery, and Matthews [8, 30] surveys it. Venturini [31, 32] studies the same maps as *periodically linear* functions, which is the terminology of our Proposition 13.2. Mignosi [33] and Brocco [34] treat a related generalisation, and Rouet and Feix [35] construct cycles and give a stochastic model. Matthews and Leigh [36] and Hicks et al. [37] move to polynomial rings, where the analogue of the conjecture is provable. Our contribution at $q \geq 3$ is only Propositions 12.1 and 12.2 and Theorem 12.5, which we have not found stated in this form, and the drift threshold $q^q$.

**Undecidability.** Conway [9] and Kurtz and Simon [10] bound the whole programme from above, as discussed above and in Section 13.

# 2 The family

**Definition 2.1.** For $a \in \mathbb{Z}_{\geq 0}$ and $b \in \mathbb{Z}$ define $T_{a,b} : \mathbb{Z} \to \mathbb{Z}$ by $T_{a,b}(n) = n/2$ for $n$ even and $T_{a,b}(n) = an + b$ for $n$ odd. We call the pair $(a, b)$ a *universe* and write it $ax + b$. The phase space is $\mathbb{Z}$; Proposition 2.2 says when each half-line is invariant, and Proposition 3.4 reduces $b < 0$ to $b > 0$, so the classification of Section 6 is stated for $b \geq 0$ without loss.

## 2.1 The general $q$-ary map

Definition 2.1 is the case $q = 2$ of a wider family. Fix a modulus $q \geq 1$ and coefficients $a_1, \ldots, a_{q-1}$, $b_1, \ldots, b_{q-1}$, and define

$$F(n) \;=\; \begin{cases} a_i n + b_i & n \equiv i \pmod{q}, \quad 1 \leq i \leq q-1, \\ n/q & n \equiv 0 \pmod{q}. \end{cases}$$

For $q = 1$ every $n$ is divisible by $q$ and $F$ is the identity, so the family is trivial. Sections 3–9 study $q = 2$, the case with a single affine branch; we write $T_{a,b}$ for it and call $a$ the *multiplier* and $b$ the *offset*.

Most of that material is specific to $q = 2$, because it uses the fact that the parity of $an + b$ is forced by those of $a$, $b$ and $n$. Some of it is not, and Section 12 returns to general $q$ to say exactly which parts survive: both symmetries do, with 2 replaced by $q$ and "odd" replaced by "coprime to $q$"; the parity obstruction becomes a statement about a finite directed graph on $\mathbb{Z}/q$; and rigidity fails outright.

## 2.2 The phase space over $\mathbb{Z}$

It is natural to take $\mathbb{N}_{\geq 1}$ as the phase space, and much of the literature does. But $T_{a,b}$ is defined on all of $\mathbb{Z}$, and restricting to the positive integers hides structure — most conspicuously the three known cycles of $3x - 1$, which are the negative-side cycles of the Collatz map itself. We therefore work over $\mathbb{Z}$ and record exactly when the sign is preserved.

**Proposition 2.2** (Structure of $\mathbb{Z}$)**.** *Let $a \geq 1$ and $b \in \mathbb{Z}$.*

*(i) $T_{a,b}(0) = 0$, and $T_{a,b}(n) = 0$ for $n \neq 0$ exactly when $n$ is odd with $n = -b/a$. So $\{0\}$ is a fixed point with at most one other preimage.*

*(ii) $\mathbb{Z}_{\geq 1}$ is $T_{a,b}$-invariant if and only if $a + b \geq 1$.*

*(iii) $\mathbb{Z}_{\leq -1}$ is $T_{a,b}$-invariant if and only if $b \leq a - 1$.*

*(iv) In all cases the sign can change only at an* odd *$n$ with $|n| \leq |b|/a$. Hence at most $\lceil |b|/a \rceil$ integers of each sign ever cross, and every orbit is eventually confined to one of $\mathbb{Z}_{\geq 1}$, $\{0\}$, $\mathbb{Z}_{\leq -1}$.*

*Proof.* (i) 0 is even and $0/2 = 0$. If $n \neq 0$ maps to 0 then $n$ cannot be even (as $n/2 = 0$ forces $n = 0$), so $n$ is odd and $an + b = 0$.

(ii) Halving preserves $\mathbb{Z}_{\geq 1}$, so invariance can fail only on the odd branch. For odd $n \geq 1$ the quantity $an + b$ is increasing in $n$, so its minimum over odd $n \geq 1$ is $a + b$, attained at $n = 1$. Thus $an + b \geq 1$ for all odd $n \geq 1$ iff $a + b \geq 1$.

(iii) Apply (ii) to $T_{a,-b}$ and use Proposition 3.4: the negatives of $T_{a,b}$ are the positives of $T_{a,-b}$, which are invariant iff $a - b \geq 1$.

(iv) Halving never changes sign, so a change of sign happens at an odd $n$. If $n \geq 1$ is odd and $an + b \leq 0$ then $an \leq -b$, so $b < 0$ and $n \leq |b|/a$; symmetrically for $n \leq -1$. That bounds the crossing set, which is therefore finite.

For the confinement, note that in each case at least one half-line is already invariant, and crossings can only run *into* it. If $b > 0$ then $a + b \geq 1$, so $\mathbb{Z}_{\geq 1}$ is invariant by (ii) and every crossing goes from negative to positive; if $b < 0$ then $b \leq a - 1$, so $\mathbb{Z}_{\leq -1}$ is invariant by (iii) and every crossing goes the other way; and if $b = 0$ then $an$ has the sign of $n$ and there are no crossings at all. In each case an orbit changes sign at most once, after which it is confined. □

**Remark 2.3.** Proposition 2.2 is why the paper can state its theorems over $\mathbb{Z}$ without a case explosion. For $b \geq 0$ — the normalisation we use after Proposition 3.4 — part (ii) holds automatically whenever $a \geq 1$, so the positive integers are invariant; part (iii) then says the negatives are invariant precisely when $b < a$, and otherwise finitely many negative integers leak into $\mathbb{Z}_{\geq 0}$. The classical case $b = 1 < 3 = a$ has both half-lines invariant, which is why $3x + 1$ and $3x - 1$ are usually treated as separate problems. They are the two halves of one map.

**Definition 2.4** (Fates)**.** An orbit is *eventually periodic* if it repeats a value, and *divergent* if $|\cdot|$ is unbounded along it. Fix an invariant set $S \subseteq \mathbb{Z}$ — in practice $\mathbb{N}_{\geq 1}$ or $\mathbb{Z}_{\leq -1}$, by Proposition 2.2. The *fate of $T_{a,b}$ on $S$* is CYCLIC if every orbit from $S$ is eventually periodic, DIVERGENT if every orbit from $S$ is divergent, and OPEN otherwise or when unknown. An unqualified "fate" means the fate on $\mathbb{N}_{\geq 1}$; by Proposition 3.4 the fate on $\mathbb{Z}_{\leq -1}$ is the fate of $T_{a,-b}$ on $\mathbb{N}_{\geq 1}$, so the two half-lines are covered by classifying $(a, b)$ and $(a, -b)$. We write $\mathrm{Cyc}(T_{a,b})$ for the set of cycles in $S$, each normalised to begin at its least element.

Two trivialities are used constantly and recorded once.

**Lemma 2.5.** *Let $n \in \mathbb{Z}$ with $n \neq 0$. Then the $T_{a,b}$-orbit of $n$ contains infinitely many odd terms, or reaches* 0*. If moreover $a \geq 1$ and $a + b \geq 1$ and $n \geq 1$, then $T^k_{a,b}(n) \geq 1$ for all $k$.*

*Proof.* Halving strictly decreases $|n|$ for $n \neq 0$, so no orbit can consist of even nonzero terms from some point on: after at most $v_2(n)$ halvings an odd value or 0 appears, and the same applies to every tail. The positivity claim is Proposition 2.2(ii). $\square$

**Lemma 2.6** (Valuations)**.** *Let $p$ be an odd prime and $n \neq 0$. Then $v_p(n/2) = v_p(n)$ whenever $n$ is even, and*

$$v_p(an+b) \;\geq\; \min(v_p(a) + v_p(n),\; v_p(b)),$$

*with equality whenever $v_p(a) + v_p(n) \neq v_p(b)$. (Here $v_p(0) := +\infty$.)*

*Proof.* The first claim holds because $p$ is odd. The second is the ultrametric inequality for $v_p$ together with $v_p(an) = v_p(a) + v_p(n)$. $\square$

# 3 The symmetries

## 3.1 Halving: carykh's even parallel universes

**Proposition 3.1** (Halving)**.** *Let $U = T_{2a,2b}$ and $T = T_{a,b}$, with $(a,b) \neq (0,0)$. Then for every $n$,*

$$n \text{ even} \implies U(n) = T(n), \qquad n \text{ odd} \implies U(n) = 2\,T(n) \text{ is even and } U^2(n) = T(n).$$

*Consequently the $U$-orbit of $n$ is obtained from the $T$-orbit of $n$ by inserting the single extra term $2\,T(x)$ immediately after each odd term $x$. In particular:*

*(i) $n$ is $U$-eventually-periodic iff it is $T$-eventually-periodic, and $U$-divergent iff $T$-divergent; the two universes have the same fate.*

*(ii) $C \mapsto C \cup \{\, 2\,T(x) : x \in C \text{ odd} \}$ is a bijection $\mathrm{Cyc}(T) \to \mathrm{Cyc}(U)$, and the corresponding periods satisfy $|C_U| = |C_T| + k(C_T)$, where $k$ counts the odd terms of $C_T$.*

*Proof.* For even $n$ both maps halve. For odd $n$, $U(n) = 2an + 2b = 2(an+b) = 2T(n)$, which is even, so $U$ halves it next and $U^2(n) = an + b = T(n)$. The description of the orbit follows by induction, and (i) is immediate because the two orbits have the same set of values — the inserted terms are determined by their predecessors — and are simultaneously finite or unbounded. For (ii), every $U$-cycle contains an odd term by Lemma 2.5, and starting from such a term the displayed identities read off the $T$-cycle; conversely each $T$-cycle inflates to a $U$-cycle. For the period: starting at $x \in C_T$, the insertion description returns to $x$ after exactly $|C_T| + k(C_T)$ steps of $U$, and no sooner — deleting the inserted terms from a shorter $U$-return would give a shorter $T$-return, contradicting the minimality of $|C_T|$. Since a periodic orbit of a map has exactly as many distinct elements as its period, $|C_U| = |C_T| + k(C_T)$, and in particular the inserted terms are automatically distinct from the elements of $C_T$. $\square$

Proposition 3.1 is carykh's observation that $4x+2$ "is just $2x+1$ with more steps" [1], made exact. Note what it is *not*: the two maps are not conjugate, because $U$ takes strictly more steps. Theorem 3.5 below shows that no conjugacy could exist.

## 3.2 Odd scaling: carykh's odd parallel universes

**Proposition 3.2** (Odd scaling)**.** *Let $d$ be an odd integer. Then for every $n$,*

$$T_{a,\,db}(dn) \;=\; d \cdot T_{a,b}(n).$$

*Hence $d\mathbb{Z}$ is $T_{a,db}$-invariant and $n \mapsto dn$ is an isomorphism of dynamical systems from $T_{a,b}$ onto $T_{a,db}$ restricted to $d\mathbb{Z}$.*

*Proof.* Since $d$ is odd, $dn$ and $n$ have the same parity. If $n$ is even, $T_{a,db}(dn) = dn/2 = d(n/2)$. If $n$ is odd, $T_{a,db}(dn) = a\,dn + db = d(an+b)$. $\square$

**Example 3.3.** $T_{5,5}$ restricted to $5\mathbb{Z}$ is a copy of $T_{5,1}$. carykh observed that the two have the identical "cycle fingerprint" of periods $7, 10, 10$; by Proposition 3.2 the cycles of $T_{5,5}$ that lie in $5\mathbb{Z}$ are exactly 5 times those of $T_{5,1}$, so this is forced. Our census confirms that *all three* cycles of $T_{5,5}$ lie in $5\mathbb{Z}$, which is a separate fact — it is Theorem 4.1(i), since $5 \mid \gcd(5,5)$.

### 3.3 Negation

**Proposition 3.4** (Negation duality)**.** *For all $a, b$ and all $n$, $T_{a,-b}(n) = -T_{a,b}(-n)$. Hence $n \mapsto -n$ conjugates $T_{a,b}$ on $\mathbb{Z}_{\le -1}$ to $T_{a,-b}$ on $\mathbb{N}_{\ge 1}$.*

*Proof.* $n$ and $-n$ have the same parity. For $n$ even both sides are $n/2$; for $n$ odd, $-T_{a,b}(-n) = -(-an+b) = an - b = T_{a,-b}(n)$. □

So the negative half of the classical Collatz map is the positive half of $3x-1$, whose three known cycles begin at 1, 5 and 17. Everything below is stated for $b \ge 0$; Proposition 3.4 transfers it.

### 3.4 Rigidity: there is nothing else

**Theorem 3.5** (Rigidity)**.** *Let $\varphi(n) = \lambda n + \mu$ with $\lambda \neq 0$, let $\varphi$ map $\mathbb{Z}$ into $\mathbb{Z}$, and suppose $\varphi \circ T_{a,b} = T_{a',b'} \circ \varphi$ on $\mathbb{Z}$, where $a, b, a', b' \in \mathbb{Z}$ and $a' \neq 0$. Then*

$$\mu = 0, \qquad \lambda \text{ is an odd integer}, \qquad a' = a, \qquad b' = \lambda b.$$

*Proof.* Taking $n = 0$ and $n = 1$ shows $\mu, \lambda + \mu \in \mathbb{Z}$, so $\lambda, \mu \in \mathbb{Z}$. We rule out the bad parities in turn. Throughout, "$\varphi(n)$ even" selects the halving branch of $T_{a',b'}$.

*Case $\lambda$ even.* Then $\varphi(n) \equiv \mu \pmod 2$ for every $n$, so $T_{a',b'} \circ \varphi$ uses one fixed branch. If $\mu$ is even, evaluating the intertwining relation at even $n$ gives $\lambda n/2 + \mu = (\lambda n + \mu)/2$, hence $\mu = 0$; then at odd $n$ it gives $\lambda(an+b) = \lambda n/2$ for all odd $n$, forcing $a = 1/2$, which is not an integer. If $\mu$ is odd, evaluating at even $n$ gives $\lambda n/2 + \mu = a'(\lambda n + \mu) + b'$ for all even $n$; comparing the coefficients of $n$ yields $\lambda/2 = a'\lambda$, i.e. $a' = 1/2$, again impossible.

*Case $\lambda$ odd, $\mu$ odd.* Then $\varphi$ reverses parity. At even $n$ the relation reads $\lambda n/2 + \mu = a'(\lambda n + \mu) + b'$, and comparing coefficients of $n$ gives $a' = 1/2$: impossible.

*Case $\lambda$ odd, $\mu$ even.* Then $\varphi$ preserves parity. At even $n$: $\lambda n/2 + \mu = (\lambda n + \mu)/2$, so $\mu = 0$. At odd $n$: $\lambda(an+b) = a'\lambda n + b'$ for all odd $n$; comparing coefficients gives $a' = a$ and $b' = \lambda b$. □

**Corollary 3.6.** *(i) The only bijective affine conjugacies within the family are the identity and $n \mapsto -n$, which gives $T_{a,b} \cong T_{a,-b}$.*

*(ii) The multiplier $a$ is an affine invariant: no affine change of variable relates two universes in different rows. In particular no such change of variable can reduce $5x+1$ to $3x+1$.*

*(iii) Within a row, two universes are affinely related only if their offsets differ by an odd factor.*

*Proof.* All three are immediate from Theorem 3.5, which gives $\mu = 0$, $\lambda$ odd, $a' = a$ and $b' = \lambda b$. For (i), a bijection of $\mathbb{Z}$ of the form $n \mapsto \lambda n$ forces $\lambda = \pm 1$, and $\lambda = -1$ gives $b' = -b$. □

Corollary 3.6(ii) is worth emphasising, because it is the precise sense in which the rows of the multiverse are independent problems. Whatever resolves $3x+1$ need not touch $5x+1$.

### 3.5 Modified maps: what generalises, and what does not

Propositions 3.1 and 3.2 both have the shape "post-compose the odd branch with a map $g$ and see what survives". It is natural to ask how far that goes. Let $g : \mathbb{Z} \to \mathbb{Z}$ be arbitrary and set

$$T_g(n) = \begin{cases} n/2 & n \text{ even}, \\ g(an+b) & n \text{ odd}. \end{cases}$$

**Proposition 3.7** (Post-composition). *For every odd $n$, $T_g(n) = g(T_{a,b}(n))$.*

*Proof.* Immediate from the definitions: both sides are $g(an + b)$. $\square$

Proposition 3.7 is genuine but shallow — it says nothing about the even branch, and it is the even branch that carries the dynamics. Taking $g(y) = 2y$ recovers $T_g = T_{2a,2b}$ and hence Proposition 3.1; taking $g(y) = cy$ gives $T_g = T_{ca,\,cb}$.

It is tempting to go further and claim that $g$ itself conjugates $T_{a,b}$ to $T_g$ whenever $g$ preserves 2-adic valuations — in particular that for odd $c$, every cycle $C$ of $T_{a,b}$ lifts to the cycle $cC$ of $T_{ca,cb}$. That is false, and it fails immediately.

**Proposition 3.8** (No such conjugacy). *Let $c$ be odd, $c > 1$. Then $g(y) = cy$ does not in general carry cycles of $T_{a,b}$ to cycles of $T_{ca,cb}$. For example $C = \{1, 4, 2\}$ is a cycle of $T_{3,1}$ and $v_2(3x) = v_2(x)$ for every $x$, yet $3C = \{3, 12, 6\}$ is not a cycle of $T_{9,3}$: indeed $T_{9,3}(3) = 9 \cdot 3 + 3 = 30 \neq 12$. More generally, a map $\varphi(n) = \lambda n$ intertwines $T_{a,b}$ with $T_{a',b'}$ only when $a' = a$ (Theorem 3.5), whereas $T_{ca,cb}$ has multiplier $ca \neq a$.*

*Proof.* The displayed computation is the counterexample. For the general statement, Theorem 3.5 forces $a' = a$ for any affine intertwining, and $ca = a$ only if $c = 1$. $\square$

An exhaustive check over the cycles of $T_{a,b}$ for $a \in \{3, 5, 7\}$, $b \in \{1, 3, 5\}$ and $c \in \{3, 5, 7, 9\}$ finds *no* case in which $cC$ is a cycle of $T_{ca,cb}$ — 0 successes out of 104 — while the same check against $T_{a,cb}$, which scales the offset only, succeeds in all 104. The correct statement is Proposition 3.2: it is the offset that may be scaled, not the pair. Rigidity (Theorem 3.5) explains why no repair is possible: $a$ is an affine invariant, so nothing that changes the multiplier can be a conjugacy.

# 4 The $p$-adic stratification

This section proves the dichotomy that organises the whole odd part of the multiverse. Fix an odd prime $p$ and write $\alpha = v_p(a)$, $\beta = v_p(b)$, with the convention $v_p(0) = +\infty$.

**Theorem 4.1** (Stratification). *Let $a \geq 1$, $b \geq 1$, and let $p$ be an odd prime. Put $\Phi(n) = \min(v_p(n), \beta)$ for $n \neq 0$. (Nothing below uses the sign of $n$, so the statement holds on all of $\mathbb{Z} \setminus \{0\}$, not merely on $\mathbb{N}_{\geq 1}$.)*

- *(i)* (Attraction.) *If $\alpha \geq 1$ then $\Phi$ is non-decreasing along every orbit, and strictly increases at every odd step at which $\Phi < \beta$. Consequently every orbit satisfies $\Phi = \beta$ — that is, lies in $p^\beta\mathbb{Z}$ — after at most $\beta$ odd steps, and remains there forever.*
- *(ii)* (Repulsion.) *If $\alpha = 0$ then $\Phi$ is* exactly *conserved: $\Phi(T_{a,b}(n)) = \Phi(n)$ for every $n \geq 1$.*

*Proof.* Halving does not change $v_p$ (Lemma 2.6), hence does not change $\Phi$; so only odd steps matter. Let $n$ be odd with $v_p(n) = k$.

(i) Let $\alpha \geq 1$. By Lemma 2.6, $v_p(an + b) \geq \min(\alpha + k, \beta) \geq \min(k + 1, \beta)$, so

$$\Phi(T_{a,b}(n)) \;\geq\; \min(\min(k+1, \beta),\, \beta) \;=\; \min(k+1, \beta).$$

If $\Phi(n) = \min(k, \beta) < \beta$ then $k < \beta$, so $\min(k + 1, \beta) > k = \Phi(n)$ and $\Phi$ strictly increases; otherwise $\Phi(n) = \beta$ and $\Phi(T_{a,b}(n)) \geq \min(k + 1, \beta) = \beta$, so $\Phi$ stays at $\beta$. Since $\Phi$ takes values in $\{0, 1, \ldots, \beta\}$ and every orbit meets odd numbers infinitely often (Lemma 2.5), the value $\beta$ is reached within $\beta$ odd steps and never left.

(ii) Let $\alpha = 0$, so $v_p(an) = k$. If $k < \beta$ then $k \neq \beta$, so Lemma 2.6 gives $v_p(an + b) = k$ exactly, whence $\Phi(T_{a,b}(n)) = k = \Phi(n)$. If $k \geq \beta$ then $v_p(an + b) \geq \min(k, \beta) = \beta$, so $\Phi(T_{a,b}(n)) = \beta = \Phi(n)$. $\square$

**Corollary 4.2** (Connected versus disconnected)**.** *Let $p$ be an odd prime dividing $b$, and let $\beta = v_p(b) \geq 1$.*

- *If $p \mid a$, the set $p^\beta\mathbb{Z} \cap \mathbb{N}_{\geq 1}$ is a global attractor: every orbit enters it and stays. On it, $T_{a,b}$ is a copy of $T_{a,\,b/p^\beta}$ by Proposition 3.2. The functional graph is that copy with a forest of transients attached.*
- *If $p \nmid a$, then $\mathbb{N}_{\geq 1}$ splits into the $\beta+1$ pairwise unreachable invariant sets $\Phi^{-1}(0), \ldots, \Phi^{-1}(\beta)$. The functional graph has at least $\beta + 1$ connected components.*

*Proof.* The first case is Theorem 4.1(i), which puts every orbit into $p^\beta\mathbb{Z}$, together with Proposition 3.2 to identify the dynamics there. The second is Theorem 4.1(ii): $\Phi$ is constant along orbits, so its level sets are invariant and no orbit joins two of them; each is non-empty because $\Phi(p^j) = \min(j, \beta)$ realises every value. □

This is precisely Artem's connected/disconnected dichotomy [3], which he states for $a = 3$ in the form "$b$ a multiple of 3 attracts, otherwise it repels". Theorem 4.1 shows the criterion is $p \mid a$ for each odd prime $p \mid b$ separately, and supplies the number of odd steps needed — $v_p(b)$ — which is Artem's observation that $3x + 3$ takes one odd step and $3x + 9$ takes two.

**Remark 4.3** ("Factor loyalty")**.** carykh notes that in $5x + 5$, once an orbit acquires a factor of 5 it never loses it, and that this happens on the whole diagonal $a = b$ when $a$ is odd. That is Theorem 4.1(i) with $\beta = 1$: for odd $a = b$, every odd prime $p \mid a$ divides both parameters, so $p\mathbb{Z}$ is entered after one odd step. He also notes correctly that the phenomenon fails on the even diagonal, e.g. $4x + 4$; there $p = 2$ is the only common prime, and Lemma 2.6 does not apply to it — halving strips powers of two, so $v_2$ is not monotone. The even diagonal instead reduces by Proposition 3.1.

# 5 Reduction and normal form

**Definition 5.1** (Rewriting rules)**.** On pairs $(a, b)$ with $a \geq 0$, $b \geq 1$ define

| | | |
|---|---|---|
| **(H)** | $(a, b) \longrightarrow (a/2,\ b/2)$ | if $2 \mid a$ and $2 \mid b$, |
| **(A)** | $(a, b) \longrightarrow (a,\ b/p^{v_p(b)})$ | if $p$ is an odd prime with $p \mid a$ and $p \mid b$. |

**Theorem 5.2** (Normal form)**.** *The rewriting system $\{$**(H)**, **(A)**$\}$ is terminating and confluent on $\{(a, b) : a \geq 0,\ b \geq 1\}$. Its normal forms are exactly the pairs with $\gcd(a, b) = 1$, so every universe has a unique normal form $\mathrm{nf}(a, b)$, and*

$$\mathrm{nf}(a, b) \;=\; \Big(\frac{a}{2^s},\ \frac{b}{2^s D}\Big), \qquad s = \min(v_2(a), v_2(b)), \quad D = \prod_{p \mid a,\ p \neq 2} p^{v_p(b)}.$$

*Proof. Termination.* Rule **(H)** decreases $\min(v_2(a), v_2(b))$ by one and leaves $v_p$ unchanged for every odd $p$; rule **(A)** removes one odd prime from $\gcd(a, b)$ — after it, $v_p(b) = 0$ — and changes no power of two, since $p^{v_p(b)}$ is odd. So the pair $\big(\min(v_2(a), v_2(b)),\ \#\{p \text{ odd prime} : p \mid \gcd(a, b)\}\big)$ strictly decreases in the product order at every step, and both coordinates are non-negative integers.

*Local confluence.* Two applicable rules act on different primes, and each rule changes only the valuation at its own prime; so any two distinct applicable steps commute. With termination, Newman's lemma gives confluence and uniqueness of normal forms.

*Normal forms are the coprime pairs.* If $\gcd(a, b) > 1$ then some prime divides both, and either $p = 2$, enabling **(H)**, or $p$ is odd, enabling **(A)**; so a normal form is coprime. Conversely neither rule applies to a coprime pair. Finally, the displayed formula is the result of applying

**(H)** exactly $s$ times and **(A)** once for each odd prime $p \mid a$ with $p \mid b$; the result is coprime, because for such a $p$ we get $v_p(b/(2^s D)) = v_p(b) - v_p(b) = 0$, and $\min(v_2(a/2^s), v_2(b/2^s)) = 0$ by choice of $s$. ☐

**Theorem 5.3** (Reduction preserves the fate)**.** *Both rules preserve the fate of a universe, and act bijectively on cycles:*

- *Under **(H)**, cycles correspond as in Proposition 3.1 (ii), with periods increased by the number of odd terms.*
- *Under **(A)** with $D = p^{v_p(b)}$, the map $C \mapsto C/D$ is a bijection $\mathrm{Cyc}(T_{a,b}) \to \mathrm{Cyc}(T_{a,b/D})$ preserving periods.*

*Consequently $T_{a,b}$ and $T_{\mathrm{nf}(a,b)}$ have the same fate.*

*Proof.* The halving case is Proposition 3.1. For **(A)**: by Theorem 4.1(i) every orbit — in particular every cycle — lies eventually, hence entirely in the case of a cycle, inside $D\mathbb{Z}$. By Proposition 3.2, $n \mapsto n/D$ carries $T_{a,b}|_{D\mathbb{Z}}$ isomorphically onto $T_{a,b/D}$, so it carries cycles to cycles bijectively and preserves periods. For fates: given $n \geq 1$, its $T_{a,b}$-orbit has a tail inside $D\mathbb{Z}$, and dividing that tail by $D$ gives a $T_{a,b/D}$-orbit; the tail is eventually periodic (resp. unbounded) iff the whole orbit is, and iff its image is. Every $m \geq 1$ arises as such an image, namely from $n = Dm$, so the universal quantifiers match. ☐

**Example 5.4.** $3x + 9 \xrightarrow{\mathbf{(A)},\, p=3} 3x + 1$: the offset loses $3^2$ at once, and Theorem 4.1(i) says the absorption into $9\mathbb{Z}$ takes two odd steps. $3x + 15 \xrightarrow{\mathbf{(A)},\, p=3} 3x + 5$, since $v_3(15) = 1$; the prime 5 is repelling and survives. $4x + 4 \xrightarrow{\mathbf{(H)}} 2x + 2 \xrightarrow{\mathbf{(H)}} 1x + 1$. All three are observations of [3] and [1], here derived.

## 5.1 One-step absorption and the radical criterion

Rule **(A)** says the offset is stripped one prime at a time, over $v_p(b)$ odd steps. There is a cruder condition under which the whole offset is stripped in a *single* step, and it is worth isolating because it is the one that is easy to test.

**Proposition 5.5** (One-step absorption)**.** *Let $a, b \geq 1$. Then $an + b \equiv 0 \pmod{b}$ for every $n$ if and only if $b \mid a$. In that case every orbit lies in $b\mathbb{Z}$ after a single odd step, and $T_{a,b}$ restricted to $b\mathbb{Z}$ is a copy of $T_{a,1}$.*

*Proof.* $an + b \equiv an \pmod{b}$, so the condition is $an \equiv 0 \pmod{b}$ for all $n$, which at $n = 1$ gives $b \mid a$; conversely if $a = kb$ then $an + b = b(kn + 1)$. The last claim is Proposition 3.2 with $d = b$. ☐

**Theorem 5.6** (Radical criterion)**.** *Let $a \geq 1$ and let $b \geq 1$ be odd. Then $\mathrm{nf}(a, b) = (a, 1)$ — equivalently, $T_{a,b}$ is, on a global attractor, a scaled copy of $T_{a,1}$ — if and only if*

$$\mathrm{rad}(b) \mid a,$$

*where $\mathrm{rad}(b) = \prod_{p \mid b} p$ is the radical of $b$.*

*Proof.* By Theorem 5.2, $\mathrm{nf}(a, b) = (a, b/D)$ with $D = \prod_{p \mid a,\, p \neq 2} p^{v_p(b)}$. Since $b$ is odd, every prime of $b$ is odd, so $b/D = 1$ iff $v_p(D) = v_p(b)$ for every $p \mid b$, iff every prime of $b$ divides $a$, iff $\mathrm{rad}(b) \mid a$. ☐

**Remark 5.7.** The two conditions differ, and the difference is exactly the phenomenon that $3x+9$ exhibits. Proposition 5.5 requires $b \mid a$, which fails for $(a,b) = (3,9)$; Theorem 5.6 requires only $\mathrm{rad}(9) = 3 \mid 3$, which holds. The absorption into $9\mathbb{Z}$ takes two odd steps rather than one, but it happens, and $3x+9$ is a scaled copy of $3x+1$. The same applies to $3x+27$, $3x+81$, $5x+25$ and $9x+27$. So $b \mid a$ is sufficient but not necessary for equivalence; the radical is the right invariant.

# 6 The solved families

We now dispose of every universe that is not primitive. Throughout, $a, b \geq 0$.

**Proposition 6.1** ($a = 0$: carykh's hub-and-spoke)**.** *Let $a = 0$. If $b = 0$, every orbit reaches the fixed point 0. If $b \geq 1$, the universe is* CYCLIC *and has exactly one cycle, namely $b \to b/2 \to \cdots \to \mathrm{odd}(b) \to b$, of period $v_2(b)+1$; every orbit enters it.*

*Proof.* For $b = 0$ every odd number maps to 0 and every even number halves toward 0. For $b \geq 1$: every orbit reaches an odd number (Lemma 2.5), which maps to $b$; from $b$ the orbit halves to $\mathrm{odd}(b)$ in $v_2(b)$ steps and returns to $b$. Any cycle must contain an odd term, and every odd term maps to $b$, so every cycle contains $b$; hence there is exactly one. □

**Proposition 6.2** ($b = 0$: carykh's shapes)**.** *Let $b = 0$ and $a \geq 1$, and write $a = 2^t u$ with $u$ odd. If $u = 1$ the universe is* CYCLIC*: every odd $n$ lies on the cycle $n \to 2^t n \to \cdots \to n$ of period $t+1 = \log_2(a)+1$, and there are infinitely many cycles. If $u \geq 3$ the universe is* DIVERGENT*.*

*Proof.* For odd $n$, $T(n) = 2^t un$ and the next $t$ steps halve it to $un$. So on odd numbers the induced map is $n \mapsto un$. If $u = 1$ this is the identity and the displayed cycle is immediate; the period is $t+1$ because one odd step is followed by exactly $t$ halvings. If $u \geq 3$ the odd terms form the strictly increasing sequence $n, un, u^2 n, \cdots \to \infty$. □

Proposition 6.2 is carykh's "all powers of two create clean-cut shapes with $\log_2(n)+1$ sides", with the accompanying observation that $3x$ and $5x$ run away and that $6x$ runs away at the reduced rate of its odd part.

**Theorem 6.3** (Parity obstruction: carykh's runaways)**.** *Let $a \geq 1$ and $b \geq 1$ with $a+b$ odd. Then every orbit from $\mathbb{N}_{\geq 1}$ diverges; indeed every orbit is eventually a strictly increasing sequence of odd numbers. The growth is linear if $a = 1$ and exponential if $a \geq 2$.*

*Proof.* If $n$ is odd then $an+b$ is odd, because exactly one of $an$, $b$ is even. So once an orbit reaches an odd number — which it does, by Lemma 2.5 — it never again reaches an even one, and thereafter $n \mapsto an+b$ with $a \geq 1$, $b \geq 1$, which strictly increases. The growth rates are those of the recursion $n_{j+1} = an_j + b$. □

Theorem 6.3 explains the checkerboard of runaway cells that dominates carykh's grid, and his "finite state machine" picture: the odd → even transition simply does not exist. His four subtypes (linear, exponential, and their halving-images) are Theorem 6.3 together with Proposition 3.1.

**Theorem 6.4** ($a = 1$: carykh's black holes)**.** *Let $a = 1$ and let $b \geq 1$ be odd. Then:*

*(i) every $n > b$ strictly decreases within at most two steps;*

*(ii) every orbit from $\mathbb{N}_{\geq 1}$ is eventually periodic;*

*(iii) every cycle has least element at most $b$, and is contained entirely in $[1, 2b]$;*

*(iv) $\{b, 2b\}$ is a cycle, and the bound $2b$ in (iii) is attained by it, so (iii) is sharp;*

(v) *over $\mathbb{Z}$: every orbit from $\mathbb{Z}_{\le -1}$ enters $\mathbb{Z}_{\ge 0}$ in finitely many steps, so* every *orbit of $T_{a,b}$ on $\mathbb{Z}$ is eventually periodic.*

*In particular the number of cycles is finite and effectively computable: it suffices to run the seeds $1, \dots, b$.*

*Proof.* (i) If $n$ is even then $T(n) = n/2 < n$. If $n$ is odd and $n > b$ then $n + b$ is even and $T^2(n) = (n + b)/2 < n$.

(ii) By (i) the orbit strictly decreases until it enters the finite set $[1, b]$, so it must repeat.

(iii) Let $C$ be a cycle and $m = \min C$. By (i), $m > b$ would make $m$ strictly decrease, contradicting minimality; so $m \le b$.

For the upper bound, list the odd elements of $C$ in cyclic order as $y_1 \to y_2 \to \cdots \to y_k \to y_1$, where $y_{i+1}$ is reached from $y_i$ by one odd step followed by $d_i \ge 1$ halvings; the list is non-empty by Lemma 2.5. Then

$$y_{i+1} \;=\; \frac{y_i + b}{2^{d_i}} \;\le\; \frac{y_i + b}{2}.$$

First, some $y_i \le b$: otherwise $y_i > b$ for every $i$, and then $y_{i+1} \le (y_i + b)/2 < y_i$ for every $i$, so the $y_i$ strictly decrease around a finite cycle, which is impossible. Second, if $y_i \le b$ then $y_{i+1} \le (b + b)/2 = b$. By induction around the cycle, therefore, $y_i \le b$ for *all* $i$.

Now every element of $C$ is either odd, hence $\le b$, or even, hence of the form $(y_i + b)/2^j$ for some $i$ and some $0 \le j < d_i$, hence at most $y_i + b \le 2b$. Thus $C \subseteq [1, 2b]$.

(iv) $T(b) = b + b = 2b$ and $T(2b) = b$, and this cycle contains $2b$.

(v) Let $n \le -1$. If $n$ is even, $|n|$ halves. If $n$ is odd, $T(n) = n + b > n$, and $|T(n)| < |n|$ whenever $n < -b$. So $|n|$ is strictly decreasing until the orbit reaches $[-b, -1]$, from which one odd step lands in $[0, b - 1] \subseteq \mathbb{Z}_{\ge 0}$. Then (ii) applies. (This is Proposition 2.2(iii) in concrete form: $b \le a - 1 = 0$ fails, so $\mathbb{Z}_{\le -1}$ is not invariant.) □

**Remark 6.5.** Part (iii) is due to the present authors' earlier notes and is strictly stronger than the bound on $\min C$ alone: it confines the whole cycle, which is what makes the census in Table 1 *exact* rather than a lower bound. A search over all odd $b \le 159$ finds $\max(C)/b \le 2$ with equality attained, confirming sharpness.

**Remark 6.6** (The $a = 1$ census is complete, and there are usually more than two cycles)**.** Theorem 6.4 makes this the one open-ended family whose cycle count is not merely a lower bound: since every cycle meets $[1, b]$, testing the seeds $1, \dots, b$ finds all of them. Table 1 is therefore exact.

It also corrects a natural guess. The orbit of 1 and the two-cycle $\{b, 2b\}$ are always present, and for small $b$ they are all there is — so one might expect exactly two cycles for every odd $b > 1$. That fails as soon as $b = 7$, which has three: $\{1, 8, 4, 2\}$, $\{3, 10, 5, 12, 6\}$ and $\{7, 14\}$. The count is 2 for $b = 11, 13, 19, 29, 37, 53, 59, 61$ but reaches 13 at $b = 63$. At $b = 1$ the two named cycles coincide, so the count is 1.

Theorem 6.4 is the sharpening of carykh's remark that the $a = 1$ row has "crushing decreases while barely giving any chance to grow". It is worth stating what it costs: the same two-step argument fails for $a = 3$ at exactly the point where $(3n + b)/2 < n$ would require $n < -b$. The $a = 1$ row is solved and the $a = 3$ row is not, and this is the whole of the difference.

## 6.1 The cycle through 1

A separate and very concrete question is when 1 itself lies on a cycle. It has a complete answer, and the answer explains a family of universes that would otherwise look accidental.

**Theorem 6.7** (Power-of-two criterion)**.** *Let $a \ge 1$ and $b \ge 1$. Then 1 lies on a cycle of $T_{a,b}$ if and only if the forward orbit of 1 contains a power of 2.*

| $b$ | all | prim | $b$ | all | prim | $b$ | all | prim | $b$ | all | prim |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 21 | 6 | 2 | 41 | 3 | 2 | 61 | 2 | 1 |
| 3 | 2 | 1 | 23 | 3 | 2 | 43 | 4 | 3 | 63 | 13 | 6 |
| 5 | 2 | 1 | 25 | 3 | 1 | 45 | 8 | 2 | 65 | 7 | 4 |
| 7 | 3 | 2 | 27 | 4 | 1 | 47 | 3 | 2 | 67 | 2 | 1 |
| 9 | 3 | 1 | 29 | 2 | 1 | 49 | 5 | 2 | 69 | 6 | 2 |
| 11 | 2 | 1 | 31 | 7 | 6 | 51 | 8 | 4 | 71 | 3 | 2 |
| 13 | 2 | 1 | 33 | 5 | 2 | 53 | 2 | 1 | 73 | 9 | 8 |
| 15 | 5 | 2 | 35 | 6 | 2 | 55 | 5 | 2 | 75 | 8 | 2 |
| 17 | 3 | 2 | 37 | 2 | 1 | 57 | 5 | 2 | 77 | 6 | 2 |
| 19 | 2 | 1 | 39 | 5 | 2 | 59 | 2 | 1 | 79 | 3 | 2 |

Table 1: Cycles of $x + b$ for odd $b \leq 79$, with the primitive count of Corollary 8.3. Unlike every other census in this paper these counts are *exact*, by Remark 6.6.

*Proof.* ($\Leftarrow$) If $T^j(1) = 2^k$ then $k$ further halvings give $T^{j+k}(1) = 1$, so 1 is periodic.

($\Rightarrow$) Suppose 1 lies on a cycle $C$ and let $y \in C$ be its predecessor. If $y$ is even then $y = 2$, a power of 2, and $y$ lies on the orbit. If $y$ is odd then $ay+b = 1$ with $y \geq 1$, forcing $a+b \leq ay+b = 1$, impossible for $a, b \geq 1$. $\square$

**Corollary 6.8** ($a + b$ a power of two). *If $a + b = 2^k$ then 1 lies on a cycle of $T_{a,b}$, of period exactly $k + 1$, namely $1 \to 2^k \to 2^{k-1} \to \cdots \to 2 \to 1$.*

*Proof.* $T(1) = a + b = 2^k$, and the remaining $k$ steps are halvings. $\square$

Corollary 6.8 is the "$a = 2^k - b$" observation of the authors' earlier notes. Theorem 6.7 shows it is the one-step case of a criterion that is both necessary and sufficient, and the gap is substantial: among the 147 pairs with $a$ odd, $a < 200$, $b$ odd, $b < 60$ for which 1 lies on a cycle, exactly 93 have $a + b$ a power of two, while the other 54 reach a power of two only after a longer excursion. For instance in $3x + 11$ the orbit of 1 runs $1 \to 14 \to 7 \to 32 \to 16 \to 8 \to 4 \to 2 \to 1$, meeting $2^5$ at the fourth step although $3 + 11 = 14$ is not a power of two.

**Remark 6.9.** Corollary 6.8 generalises verbatim to the $q$-ary map of Section 2.1: if $a + b = q^k$ then 1 lies on a cycle of period $k + 1$. Theorem 6.7 generalises too, with "power of 2" replaced by "power of $q$".

We can now state the classification.

**Theorem 6.10** (Master classification). *Let $a \geq 0$ and $b \geq 1$, and let $(a', b') = \mathrm{nf}(a, b)$. Then $T_{a,b}$ has the same fate as $T_{a',b'}$ (Theorem 5.3), and exactly one of the following holds.*

1. *$a' = 0$. The fate is* CYCLIC, *with a single cycle of period $v_2(b') + 1$ (Proposition 6.1).*
2. *$a' \geq 1$ and $a' + b'$ is odd. The fate is* DIVERGENT *(Theorem 6.3).*
3. *$a' = 1$ and $b'$ is odd. The fate is* CYCLIC, *with every cycle contained in $[1, 2b']$, so the cycle set is finite and computable (Theorem 6.4).*
4. *$a' \geq 3$ is odd, $b'$ is odd, and $\gcd(a', b') = 1$. The fate is* OPEN. *We call such a universe* primitive.

*The case $b = 0$ is separate and is settled by Proposition 6.2: the fate is* CYCLIC *if $a$ is a power of two (including $a = 1$),* DIVERGENT *otherwise, and $a = b = 0$ is terminal.*

*Proof.* Exhaustiveness. By Theorem 5.2, $\gcd(a', b') = 1$, so $a'$ and $b'$ are not both even. If $a' = 0$ we are in (1). Otherwise $a' \geq 1$; if $a' + b'$ is odd we are in (2). If $a' + b'$ is even and they are not both even, they are both odd; then $a' = 1$ gives (3) and $a' \geq 3$ gives (4). Mutual exclusivity is clear. The cited results supply the fates. $\square$

**Remark 6.11.** Nothing in Theorem 6.10 is a probabilistic or heuristic statement: cases (1)–(3) are proved outright, and case (4) is a statement that a proof is not known, not a statement that the fate is mixed. What *is* heuristic is which fate case (4) is expected to have; see Section 9.

# 7 The table of universes

Table 2 classifies every universe with $0 \le a, b \le 12$ according to Theorem 6.10. The codes are **C** (constant, $a = 0$), **S** (shape, $b = 0$ with $a$ a power of two), **R** (runaway — all orbits diverge), **H** (halving-reducible), **A** (attraction-reducible), **D** ($a = 1$ descent) and **?** (primitive and open).

| $a\backslash b$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | **C** | **C** | **C** | **C** | **C** | **C** | **C** | **C** | **C** | **C** | **C** | **C** | **C** |
| 1 | **S** | **D** | **R** | **D** | **R** | **D** | **R** | **D** | **R** | **D** | **R** | **D** | **R** |
| 2 | **S** | **R** | **H** | **R** | **H** | **R** | **H** | **R** | **H** | **R** | **H** | **R** | **H** |
| 3 | **R** | **?** | **R** | **A** | **R** | **?** | **R** | **?** | **R** | **A** | **R** | **?** | **R** |
| 4 | **S** | **R** | **H** | **R** | **H** | **R** | **H** | **R** | **H** | **R** | **H** | **R** | **H** |
| 5 | **R** | **?** | **R** | **?** | **R** | **A** | **R** | **?** | **R** | **?** | **R** | **?** | **R** |
| 6 | **R** | **R** | **H** | **R** | **H** | **R** | **H** | **R** | **H** | **R** | **H** | **R** | **H** |
| 7 | **R** | **?** | **R** | **?** | **R** | **?** | **R** | **A** | **R** | **?** | **R** | **?** | **R** |
| 8 | **S** | **R** | **H** | **R** | **H** | **R** | **H** | **R** | **H** | **R** | **H** | **R** | **H** |
| 9 | **R** | **?** | **R** | **A** | **R** | **?** | **R** | **?** | **R** | **A** | **R** | **?** | **R** |
| 10 | **R** | **R** | **H** | **R** | **H** | **R** | **H** | **R** | **H** | **R** | **H** | **R** | **H** |
| 11 | **R** | **?** | **R** | **?** | **R** | **?** | **R** | **?** | **R** | **?** | **R** | **A** | **R** |
| 12 | **R** | **R** | **H** | **R** | **H** | **R** | **H** | **R** | **H** | **R** | **H** | **R** | **H** |

Table 2: The multiverse for $0 \le a, b \le 12$. Only the cells marked **?** are open, and only those; every other cell is settled by Theorem 6.10. The cells marked **H** and **A** are open or solved according to the fate of their normal form.

The visible regularities all have proofs above. The row $a = 0$ and the column $b = 0$ are Propositions 6.1 and 6.2. The checkerboard of **R** is Theorem 6.3. The quarter-lattice of **H** at even $(a, b)$ is Proposition 3.1. The **A** cells lie where an odd prime divides both coordinates — so the diagonal $a = b$ with $a$ odd, and more generally the sublattices $p \mid a$, $p \mid b$. What is left is the odd–odd sublattice with $\gcd = 1$.

## 7.1 carykh's nine

carykh singles out nine cells of his $6 \times 6$ grid as having "no obvious indicators" — these are exactly the cells with $a, b \in \{1, 3, 5\}$, i.e. the odd–odd sublattice, which is the correct set to single out. Theorem 6.10 resolves five of the nine.

**Theorem 7.1.** *Of carykh's nine universes, three are solved, two reduce, and four are open and pairwise affinely inequivalent:*

| *universe* | *family* | *status* |
|---|---|---|
| $1x+1$ | *descent* | SOLVED — *every orbit cycles* |
| $1x+3$ | *descent* | SOLVED — *every orbit cycles* |
| $1x+5$ | *descent* | SOLVED — *every orbit cycles* |
| $3x+1$ | *primitive* | OPEN |
| $3x+3$ | *attracting-reducible* | REDUCES — *equivalent to* $3x+1$ |
| $3x+5$ | *primitive* | OPEN |
| $5x+1$ | *primitive* | OPEN |
| $5x+3$ | *primitive* | OPEN |
| $5x+5$ | *attracting-reducible* | REDUCES — *equivalent to* $5x+1$ |

*The four open universes are* $3x+1$, $3x+5$, $5x+1$ *and* $5x+3$.

*Proof.* $1x+1$, $1x+3$, $1x+5$ are Theorem 6.4. $3x+3$ and $5x+5$ have $3 \mid \gcd(3,3)$ and $5 \mid \gcd(5,5)$, so rule **(A)** applies and Theorem 5.3 identifies their fates with those of $3x+1$ and $5x+1$. The remaining four are primitive by inspection of gcd, hence case (4) of Theorem 6.10. For pairwise inequivalence, suppose an affine $\varphi$ intertwines two of them. By Theorem 3.5 it has $a' = a$ and $b' = \lambda b$ with $\lambda$ an odd integer, and if $\varphi$ is a bijection of $\mathbb{Z}$ then $\lambda = \pm 1$. Different multipliers rule out every pair with $a = 3$ against $a = 5$; and within a row, $b' = \pm b$ fails for $(1,5)$ and for $(1,3)$. So no two of the four are affinely conjugate. $\square$

**Remark 7.2.** The four are inequivalent *as members of this family under affine change of variable*, which is what Theorem 3.5 controls. That is weaker than being non-isomorphic as abstract dynamical systems, and we do not claim the stronger statement: $n \mapsto 5n$ does embed $T_{3,1}$ into $T_{3,5}$ (Proposition 3.2), just not onto it.

**Remark 7.3.** $3x+3$ deserves a word, because carykh treats it as a distinct universe with its own character, noting that its graph has many extra branches that $3x+1$ lacks. Both observations are right and compatible: Theorem 5.3 says the two have identical fates and period-preserving matching cycles, while Corollary 4.2 says the copy of $3x+1$ sits inside $3x+3$ as an attractor with a transient forest — the branches — hanging off it. Dynamically they are the same problem; as labelled graphs they are not isomorphic.

## 7.2 How much of the multiverse is open

**Theorem 7.4** (Densities)**.** *As* $N \to \infty$,

$$\frac{\#\{(a,b) \in [1,N]^2 : T_{a,b} \text{ open}\}}{N^2} \longrightarrow \frac{1}{3},$$
$$\frac{\#\{(a,b) \in [1,N]^2 : T_{a,b} \text{ open and primitive}\}}{N^2} \longrightarrow \frac{2}{\pi^2}.$$

*Proof.* By Theorem 6.10, $T_{a,b}$ is open iff its normal form is primitive, i.e. iff $a' \geq 3$ is odd. Rule **(A)** changes neither $a$ nor the parity of $b$, and rule **(H)** is applied exactly $s = \min(v_2(a), v_2(b))$ times, so $a' = a/2^s$ and $b' = b/(2^s D)$ with $D$ odd. Both $a'$ and $b'$ are odd exactly when $v_2(a) = v_2(b)$. Hence

$$T_{a,b} \text{ open} \iff v_2(a) = v_2(b) \quad\text{and}\quad a/2^{v_2(a)} \geq 3.$$

The second condition fails only when $a$ is a power of two, a set of density zero. The natural density of $\{a : v_2(a) = s\}$ is $2^{-s-1}$, and the two coordinates are independent, so $\sum_{s\geq 0} 4^{-s-1} = \frac{1}{4}\cdot\frac{4}{3} = \frac{1}{3}$.

For the second limit: a universe is open and primitive iff $a \geq 3$ is odd, $b$ is odd and $\gcd(a,b) = 1$. Conditioning on both being odd (density $1/4$), the density of coprimality among odd pairs is

$$\prod_{p \text{ odd prime}} \left(1 - \frac{1}{p^2}\right) = \frac{\zeta(2)^{-1}}{1 - 2^{-2}} = \frac{6/\pi^2}{3/4} = \frac{8}{\pi^2},$$

and removing the density-zero set of powers of two leaves $\frac{1}{4} \cdot \frac{8}{\pi^2} = 2/\pi^2$. $\square$

| $N$ | open | open/$N^2$ | open primitive | primitive/$N^2$ |
|---:|---:|---:|---:|---:|
| 10 | 26 | 0.260000 | 14 | 0.140000 |
| 50 | 794 | 0.317600 | 486 | 0.194400 |
| 100 | 3244 | 0.324400 | 1957 | 0.195700 |
| 250 | 20646 | 0.330336 | 12508 | 0.200128 |
| 500 | 82896 | 0.331584 | 50451 | 0.201804 |
| 1000 | 332396 | 0.332396 | 202161 | 0.202161 |
| 2000 | 1331396 | 0.332849 | 809865 | 0.202466 |

Table 3: Convergence to the constants of Theorem 7.4: $1/3 = 0.3333\ldots$ and $2/\pi^2 = 0.202642\ldots$

Table 3 shows the convergence. So two thirds of the multiverse is settled outright, and of the remaining third, about 61% $(= 3 \cdot 2/\pi^2)$ is irreducibly new while the rest reduces to a smaller open universe. carykh's intuition that "Collatz picked the only rule set that wasn't obvious in some way" is a little too strong: the open region has positive density. What is true is that he picked the smallest interesting one — $3x + 1$ is the least primitive universe in the lexicographic order on $(a, b)$ with $a \geq 3$.

### 7.3 The knowledge ladder

Theorem 6.10 sorts universes by *how* they are settled. For deciding what is worth studying, a coarser grading is more useful, and it separates two things the family classification conflates. Consider $2x + 2$ and $6x + 2$: both are halving-reducible, so the classification of Section 7 gives them the same label. But $2x + 2$ reduces to $1x + 1$, which is solved, while $6x + 2$ reduces to $3x + 1$ and is therefore exactly as open as the Collatz conjecture. Any presentation that colours the two alike is misleading.

**Definition 7.5** (Knowledge tiers)**.** Write $(a', b') = \mathrm{nf}(a, b)$. The universe $T_{a,b}$ is

- CLOSED if $T_{a',b'}$ has a proved fate and its complete cycle set is given in closed form — that is, $a' = 0$, or $b' = 0$, or $a' + b'$ is odd;
- DECIDED if $a' = 1$: the fate is proved and the cycle set is finite and computable for each $b'$, but no formula in $b'$ is known;
- REDUCIBLE if $T_{a',b'}$ is open and $(a, b) \neq (a', b')$;
- OPEN if $T_{a',b'}$ is open and $(a, b) = (a', b')$.

**Proposition 7.6.** *The four tiers partition the family.* DECIDED *is exactly the set of* $(a, b)$ *with* $v_2(a) = v_2(b)$ *and* $a$ *a power of two;* OPEN *is exactly the set of primitive universes. Their natural densities are*

$$\textit{CLOSED} \to \frac{2}{3}, \qquad \textit{DECIDED} \to 0, \qquad \textit{REDUCIBLE} \to \frac{1}{3} - \frac{2}{\pi^2}, \qquad \textit{OPEN} \to \frac{2}{\pi^2}.$$

*Proof.* The tiers are mutually exclusive and exhaustive by construction, since $T_{a',b'}$ is either open or proved, and in the proved case Theorem 6.10 leaves exactly the four normal-form shapes listed. For DECIDED: $a' = 1$ means $a/2^s = 1$ with $s = \min(v_2(a), v_2(b))$, i.e. $a = 2^s$, and $b'$ odd forces $v_2(b) = s$. The powers of two have density 0, so the tier does. OPEN is the primitive set, of density $2/\pi^2$ by Theorem 7.4; the open universes have density $1/3$, so REDUCIBLE takes the difference; and the proved universes make up the remaining $2/3$, all but a density-zero part of which is CLOSED. $\square$

Table 4 confirms the four limits numerically.

| $N$ | CLOSED | DECIDED | REDUCIBLE | OPEN |
|---|---|---|---|---|
| 10 | 0.640000 | 0.100000 | 0.120000 | 0.140000 |
| 50 | 0.662400 | 0.020000 | 0.123200 | 0.194400 |
| 100 | 0.665600 | 0.010000 | 0.128700 | 0.195700 |
| 250 | 0.665664 | 0.004000 | 0.130208 | 0.200128 |
| 500 | 0.666416 | 0.002000 | 0.129780 | 0.201804 |
| 1000 | 0.666604 | 0.001000 | 0.130235 | 0.202161 |
| limit | 0.666667 | 0.000000 | 0.130691 | 0.202642 |

Table 4: Proportion of $(a, b) \in [1, N]^2$ in each tier of Definition 7.5, against the limits of Proposition 7.6.

**Remark 7.7.** The DECIDED tier is not empty, which is worth saying because it is easy to assume that a proved fate comes with a known cycle structure. The $a = 1$ row is a counterexample: Theorem 6.4 settles the fate in two lines, and Remark 6.6 shows the cycle counts $1, 2, 2, 3, 3, 2, 2, 5, 3, 2, 6, \ldots$ follow no pattern we can see. It is also the only such row: every other proved family has its cycles in closed form.

# 8 Cycles

## 8.1 The cycle equation

**Theorem 8.1** (Cycle equation)**.** *Let $C$ be a cycle of $T_{a,b}$ containing $k \geq 1$ odd terms $n_1, \ldots, n_k$ in cyclic order, where $n_{i+1}$ is reached from $n_i$ by one odd step followed by $d_i \geq 1$ halvings, and let $m = d_1 + \cdots + d_k$ be the total number of halvings. Then*

$$n_1(2^m - a^k) \ = \ b \sum_{i=1}^{k} a^{\,k-i}\, 2^{\,d_1+\cdots+d_{i-1}}.$$

*In particular $2^m > a^k$ for any cycle in $\mathbb{N}_{\geq 1}$ with $b \geq 1$, so $m/k > \log_2 a$.*

*Proof.* By construction $2^{d_i} n_{i+1} = a n_i + b$ for each $i$, indices modulo $k$. Multiply the $i$-th relation by $a^{k-i} 2^{d_1+\cdots+d_{i-1}}$ and sum over $i$; all terms $n_2, \ldots, n_k$ telescope away, leaving $2^m n_1 = a^k n_1 + b \sum_i a^{k-i} 2^{d_1+\cdots+d_{i-1}}$. The right-hand sum is positive when $b \geq 1$, and $n_1 \geq 1$, so $2^m - a^k > 0$. □

The inequality $m/k > \log_2 a$ is the exact form of carykh's remark that his two period-8 cycles in $3x + 5$ have "three rises and five falls, which corresponds to $3^3 = 27$ being close to $2^5 = 32$", and that the period-44 cycles have $k = 17$, $m = 27$ with $3^{17} = 129{,}140{,}163$ close to $2^{27} = 134{,}217{,}728$. Our census confirms both $(k, m)$ pairs exactly. The mechanism is now visible: small cycle elements require $2^m - a^k$ to be large relative to the right-hand side, but $m/k$ must exceed $\log_2 a$ only slightly for the sum to stay small — so cycles cluster at convergents of the continued fraction of $\log_2 a$. For $a = 3$, $\log_2 3 = 1.5849\ldots$ and $27/17 = 1.5882\ldots$

## 8.2 Strata and the divisor decomposition

**Theorem 8.2** (Stratification of the phase space)**.** *Let* $\gcd(a, b) = 1$ *with $a, b \geq 1$ odd. For $n \geq 1$ put*

$$\pi(n) \ = \ \prod_{p \mid b} p^{\min(v_p(n),\, v_p(b))},$$

*a divisor of $b$. Then $\pi$ is constant along orbits, and*

$$\mathbb{N}_{\geq 1} \ = \ \bigsqcup_{d \mid b} \Sigma_d, \qquad \Sigma_d = \pi^{-1}(d),$$

*is a partition into $\tau(b)$ non-empty $T_{a,b}$-invariant sets. Moreover $n \mapsto n/d$ is an isomorphism from $T_{a,b}|_{\Sigma_d}$ onto $T_{a,b/d}|_{\Sigma_1}$, where $\Sigma_1$ is the corresponding* primitive stratum *of $T_{a,b/d}$.*

*Proof.* Since $\gcd(a, b) = 1$, every prime $p \mid b$ is odd and satisfies $p \nmid a$, so Theorem 4.1(ii) applies to each: $\min(v_p(\cdot), v_p(b))$ is conserved. Hence so is $\pi$, and the fibres partition $\mathbb{N}_{\geq 1}$ into invariant sets. The value set is all of $\{d : d \mid b\}$ because $\pi(d) = d$ for each divisor $d$ of $b$: indeed for $p \mid b$ we have $\min(v_p(d), v_p(b)) = v_p(d)$.

Fix $d \mid b$ and $n \in \Sigma_d$. For each $p \mid b$, either $v_p(d) < v_p(b)$, in which case $v_p(n) = v_p(d)$ so $v_p(n/d) = 0$; or $v_p(d) = v_p(b)$, in which case $p \nmid b/d$. Either way $\pi_{b/d}(n/d) = 1$, so $n/d \in \Sigma_1$ for $T_{a,b/d}$; and the map is onto with inverse $m \mapsto dm$, which lands in $\Sigma_d$ by the same computation. Since $d$ is odd, Proposition 3.2 makes it a conjugacy. □

**Corollary 8.3** (Divisor decomposition of the cycle set)**.** *Under the hypotheses of Theorem 8.2, writing* Cyc$^*$ *for the set of cycles lying in the primitive stratum,*

$$\mathrm{Cyc}(T_{a,b}) \;=\; \bigsqcup_{d \mid b} d \cdot \mathrm{Cyc}^*(T_{a,\,b/d}),$$

*and the bijections preserve periods and the pair $(k, m)$.*

*Proof.* Each cycle lies in a single stratum, and Theorem 8.2 identifies the $d$-th stratum with the primitive stratum of $T_{a,b/d}$ scaled by $d$. Conjugacies preserve periods, and they preserve parities — $d$ is odd — hence $k$ and $m$. □

Corollary 8.3 is the precise form of Artem's statement that "a composite universe contains a parallel universe of every single one of its divisors". Two things are added. First, the copies do not merely exist: they *exhaust* the cycle set, so the count is an exact sum, not a lower bound. Second, the same holds for the whole phase space, not only the cycles (Theorem 8.2).

| universe | cycles | decomposition |
|---|---|---|
| $3x + 5$ | 6 | $1{\cdot}\mathrm{P}(3x+5)^5 \sqcup 5{\cdot}\mathrm{P}(3x+1)^1$ |
| $3x + 7$ | 2 | $1{\cdot}\mathrm{P}(3x+7)^1 \sqcup 7{\cdot}\mathrm{P}(3x+1)^1$ |
| $3x + 25$ | 8 | $1{\cdot}\mathrm{P}(3x+25)^2 \sqcup 5{\cdot}\mathrm{P}(3x+5)^5 \sqcup 25{\cdot}\mathrm{P}(3x+1)^1$ |
| $3x + 35$ | 9 | $1{\cdot}\mathrm{P}(3x+35)^2 \sqcup 5{\cdot}\mathrm{P}(3x+7)^1 \sqcup 7{\cdot}\mathrm{P}(3x+5)^5 \sqcup 35{\cdot}\mathrm{P}(3x+1)^1$ |
| $3x + 49$ | 3 | $1{\cdot}\mathrm{P}(3x+49)^1 \sqcup 7{\cdot}\mathrm{P}(3x+7)^1 \sqcup 49{\cdot}\mathrm{P}(3x+1)^1$ |
| $3x + 65$ | 16 | $1{\cdot}\mathrm{P}(3x+65)^1 \sqcup 5{\cdot}\mathrm{P}(3x+13)^9 \sqcup 13{\cdot}\mathrm{P}(3x+5)^5 \sqcup 65{\cdot}\mathrm{P}(3x+1)^1$ |

Table 5: Corollary 8.3 in action; $\mathrm{P}(ax + b)^j$ denotes the $j$ primitive cycles of $ax + b$. Compare Artem's counts of 9, 8 and 3 for $b = 35, 25, 49$.

Table 5 works the decomposition through for several offsets.

**Example 8.4** ($b = 35$)**.** $\mathrm{Cyc}(T_{3,35})$ has nine elements: two primitive, one equal to $5 \cdot \mathrm{Cyc}^*(T_{3,7})$, five equal to $7 \cdot \mathrm{Cyc}^*(T_{3,5})$, and one equal to $35 \cdot \mathrm{Cyc}^*(T_{3,1})$. Artem obtains the same nine by hand.

**Example 8.5** ($b = 65$)**.** Artem observes that multiplying two offsets with unusually many cycles gives one with even more. Corollary 8.3 makes this exact: 5 and 13 contribute 5 and 9 primitive cycles respectively, and $65 = 5 \cdot 13$ has $1 + 5 + 9 + 1 = 16$.

## 8.3 Census

Table 6 gives cycle counts for the row $a = 3$. The search window is seeds $1 \leq n \leq 20{,}000$ with a step budget of 200,000; this is a *converged* window in the sense that every count is identical at 20,000 and at 100,000 seeds. Counts remain lower bounds in principle — a cycle whose basin of

| $b$ | all | prim | $b$ | all | prim | $b$ | all | prim | $b$ | all | prim |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 27 | 1 | 1 | 53 | 2 | 1 | 79 | 5 | 4 |
| 3 | 1 | 1 | 29 | 5 | 4 | 55 | 11 | 3 | 81 | 1 | 1 |
| 5 | 6 | 5 | 31 | 2 | 1 | 57 | 2 | 1 | 83 | 4 | 3 |
| 7 | 2 | 1 | 33 | 3 | 2 | 59 | 8 | 7 | 85 | 9 | 1 |
| 9 | 1 | 1 | 35 | 9 | 2 | 61 | 3 | 2 | 87 | 5 | 4 |
| 11 | 3 | 2 | 37 | 4 | 3 | 63 | 2 | 1 | 89 | 2 | 1 |
| 13 | 10 | 9 | 39 | 10 | 9 | 65 | 16 | 1 | 91 | 14 | 3 |
| 15 | 6 | 5 | 41 | 2 | 1 | 67 | 2 | 1 | 93 | 2 | 1 |
| 17 | 3 | 2 | 43 | 2 | 1 | 69 | 4 | 3 | 95 | 10 | 3 |
| 19 | 2 | 1 | 45 | 6 | 5 | 71 | 8 | 7 | 97 | 3 | 2 |
| 21 | 2 | 1 | 47 | 8 | 7 | 73 | 4 | 3 | 99 | 3 | 2 |
| 23 | 4 | 3 | 49 | 3 | 1 | 75 | 8 | 2 | | | |
| 25 | 8 | 2 | 51 | 3 | 2 | 77 | 5 | 1 | | | |

Table 6: Cycles of $3x + b$ for odd $b \le 99$: total count, and the count of *primitive* cycles (those in the stratum $\Sigma_1$). By Corollary 8.3 the total is the sum of the primitive counts of $3x + (b/d)$ over the divisors $d \mid b$.

attraction misses the seed window is invisible to any such search — so they should be read as "at least".

Three of Artem's observations about this row are confirmed, and one is explained.

**Corollary 8.6.** *$3x + b$ has exactly one cycle whenever $b = 3^j$, and more generally $\#\mathrm{Cyc}(3x + 3^j b_0) = \#\mathrm{Cyc}(3x + b_0)$ for every $j \ge 0$ and every $b_0$ coprime to 3.*

*Proof.* $3 \mid a$, so rule **(A)** strips $3^{v_3(b)}$ entirely and Theorem 5.3 gives a period-preserving bijection of cycle sets. For $b = 3^j$ the normal form is $3x + 1$, whose only known cycle is $\{1, 4, 2\}$. $\square$

This is Artem's "only the universes that are a power of three away from the original have one loop, and the same for any prime universe". Note the statement is conditional on $3x + 1$ having only one cycle, which is not known; what *is* unconditional is the equality of counts.

Artem also observes that no $b < 100$ yields exactly seven cycles. Our converged search confirms this: the multiset of counts for odd $b \le 99$ contains $1, 2, 3, 4, 5, 6, 8, 9, 10, 11, 14$ and $16$, but not 7. We have no explanation, and we note it is sensitive to the search window — at 3,000 seeds $b = 71$ appears to have seven, and only widening the window to 20,000 reveals its eighth cycle.

Table 7 gives the complementary slice: the column $b = 1$, where $\gcd(a, 1) = 1$ forces every universe to be primitive and irreducible.

| $a$ | cycles | drift $a/4$ | periods |
|---|---|---|---|
| 1 | 1 | 0.25 | 2 |
| 3 | 1 | 0.75 | 3 |
| 5 | 3 | 1.25 | 7, 10, 10 |
| 7 | 1 | 1.75 | 4 |
| 9 | 0 | 2.25 | — |
| 11 | 0 | 2.75 | — |
| 13 | 0 | 3.25 | — |
| 15 | 1 | 3.75 | 5 |
| 17 | 0 | 4.25 | — |
| 19 | 0 | 4.75 | — |
| 21 | 0 | 5.25 | — |
| 23 | 0 | 5.75 | — |
| 25 | 0 | 6.25 | — |
| 27 | 0 | 6.75 | — |
| 29 | 0 | 7.25 | — |
| 31 | 1 | 7.75 | 6 |
| 33 | 0 | 8.25 | — |
| 35 | 0 | 8.75 | — |
| 37 | 0 | 9.25 | — |
| 39 | 0 | 9.75 | — |

Table 7: Cycles of $ax + 1$ for odd $a \leq 39$. Every entry with $a \geq 3$ is open. As the drift grows the cycles thin out and the seed window finds fewer of them.

# 9 The open region

Nothing in Section 6 distinguishes $3x + 1$ from $5x + 1$; both are case (4). The distinction is heuristic, and it is the one carykh draws when he calls the $a = 3$ row "Goldilocks" and the $a = 5$ row "Icarus".

**Definition 9.1** (Drift)**.** For a primitive universe define

$$\delta_{\text{step}} = \frac{\sqrt{a}}{2}, \qquad \delta_{\text{odd}} = \frac{a}{4}.$$

The first is the geometric mean of the two branch multipliers $a$ and $1/2$ weighted equally — carykh's "gravity". The second is the expected ratio between consecutive odd terms: an odd $n$ maps to $an + b \approx an$, which is divided by $2^v$ where $v = v_2(an + b)$ has expectation 2 under the standard equidistribution heuristic. Both cross 1 at $a = 4$, so:

- $a = 1$: $\delta_{\text{odd}} = 1/4$. Strong contraction — and here the heuristic is a theorem (Theorem 6.4).
- $a = 3$: $\delta_{\text{odd}} = 3/4$. Weak contraction. Almost all orbits are expected to be eventually periodic, and the number of cycles is expected to be finite but is not known to be.
- $a \geq 5$: $\delta_{\text{odd}} \geq 5/4$. Expansion. Almost all orbits are expected to diverge, with finitely many cycles surviving as exceptional sets.

These are heuristics with no proof in either direction, and the reader should treat the word "expected" strictly. What is worth recording is that the threshold $a < 4$ admits only $a \in \{1, 3\}$ among odd multipliers, and $a = 1$ is solved. *The entire difficulty of the family is concentrated in the single row $a = 3$, plus the expanding rows where the difficulty is of the opposite kind.*

| $a\backslash b$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|---|
| 0 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| 1 | 1.00 | 1.00 | RUN | 1.00 | RUN | 1.00 | RUN | 1.01 | RUN |
| 2 | 1.41 | RUN | 1.42 | RUN | RUN | RUN | 1.42 | RUN | RUN |
| 3 | RUN | 4.55 | RUN | 5.29 | RUN | 3.95 | RUN | 3.89 | RUN |
| 4 | 2.38 | RUN | RUN | RUN | 2.38 | RUN | RUN | RUN | RUN |
| 5 | RUN | 18.33 | RUN | 7.64 | RUN | 12.97 | RUN | 16.12 | RUN |
| 6 | RUN | RUN | 8.61 | RUN | RUN | RUN | 9.89 | RUN | RUN |
| 7 | RUN | 2.38 | RUN | 2.22 | RUN | 7.66 | RUN | 3.07 | RUN |
| 8 | 4.36 | RUN | RUN | RUN | RUN | RUN | RUN | RUN | 4.38 |

Table 8: Geometric mean of $\max(\mathrm{orbit}(n))/n$ over $n \leq 5000$; RUN marks universes where every seed exceeded the value or step budget.

### 9.1 Empirical growth statistics

Table 8 reproduces carykh's "average max growth" heat map: the geometric mean over seeds $n \leq 5000$ of $\max(\mathrm{orbit}(n))/n$.

The RUN pattern is the checkerboard of Theorem 6.3 together with its halving images. The column $b = 0$ is exactly computable, which pins down the corner of the table carykh reasons about.

**Proposition 9.2.** *Let $b = 0$ and $a = 2^t$ with $t \geq 0$. For $n \geq 1$ the relative peak is $\max(\mathrm{orbit}(n))/n = 2^{\max(v_2(n),\,t)-v_2(n)}$, and its geometric mean over $n$ — weighting $v_2(n) = j$ by its natural density $2^{-j-1}$ — is exactly*

$$2^{\,t-1+2^{-t}}.$$

*Proof.* Write $n = 2^j m$ with $m$ odd. By Proposition 6.2 the orbit descends to $m$ and then cycles through $2^t m, 2^{t-1}m, \ldots, m$, so its maximum is $\max(2^j, 2^t)\, m$ and the relative peak is $2^{\max(j,t)-j}$. The natural density of $\{n : v_2(n) = j\}$ is $2^{-j-1}$, so the mean of $\log_2$ of the relative peak is

$$\sum_{j\geq 0} 2^{-j-1}(\max(j,t) - j) = \sum_{j=0}^{t-1}(t-j)\, 2^{-j-1} = t - 1 + 2^{-t},$$

the last equality by the standard evaluation of $\sum_{j\geq 0} jx^j$ at $x = 1/2$. □

For $t = 1, 2, 3$ this gives 1.414214, 2.378414, 4.362031; the measured values over $n \leq 200{,}000$ agree to six decimal places. carykh reports $\sqrt{2}$ for the $a = 2$ row, which is the case $t = 1$; his argument — that an odd $n$ peaks at $2n$ and an even $n$ does not grow at all — is exactly the $j = 0$ term of the sum, and it is complete for $t = 1$ because no other term contributes. It does not extend: for $t \geq 2$ an even seed with $v_2(n) < t$ *does* grow, which is why the higher rows sit at $2^{\,t-1+2^{-t}}$ rather than the $2^{t/2}$ a naive reading would give. Away from $b = 0$ the $a = 2$ row is only approximately $\sqrt{2}$ — Table 8 gives 1.414, 1.417, 1.421 at $b = 0, 2, 6$ — and carykh notes this drift himself.

**Remark 9.3** (carykh's missing cell)**.** carykh reports a single anomalous blank at $6x + 34$ in his heat map and correctly diagnoses it: a seed in $3x + 17$ peaks near $5.06 \times 10^{12}$, so its image in the parallel universe $6x + 34$ peaks just above his $10^{13}$ cut-off and is misclassified as a runaway. This is an artefact of the threshold, not of the mathematics; by Proposition 3.1 the two cells must agree. It is a good illustration of why the tables here report their budgets explicitly.

## 10 Software

**`Multiverse/multiverse.py`.** A dependency-free library implementing the theory above. Its entry points map one-to-one onto the results:

| function | implements |
|---|---|
| `classify` | Theorem 6.10: family, fate, drifts, reduction chain |
| `normal_form` | Theorem 5.2 |
| `attracting_primes` / `repelling_primes` | Theorem 4.1 |
| `find_cycles` | the memoised cycle census |
| `decompose_cycles` | Corollary 8.3 |
| `cycle_equation_residual` | Theorem 8.1 |
| `growth_statistics` | Table 8, Prop. 9.2 |

A command-line interface exposes `report`, `grid` and `stats`.

```
$ python3 multiverse.py report 3 15
family      : attracting-reducible
fate        : open
normal form : 3x+5
   attraction  3x+15 -> 3x+5   (every orbit enters 3Z after at most 1 odd
       step)
attracting  : 3^1
repelling   : 5^1   (state space splits into mutually unreachable strata)
```

**`Multiverse/qary.py`.** The $q$-ary maps of Section 12 and the predicate maps of Section 13: `QMap` with its two symmetries (`scaled`, `offset_scaled`), `residue_graph` and `escaping_residues` for Theorem 12.5, `qary_drift` for Heuristic 12.7, and `primality_map` for the experiment of Proposition 13.3.

**`Multiverse/literature.py`.** Classifies the ccchallenge.org snapshot into the domains of Appendix A and emits the appendix tables. Run with `--refresh` it re-downloads the catalogue.

**`Multiverse/generate_data.py` and `inline_tables.py`.** The first regenerates every table into `Multiverse/data/` as plain text and LaTeX and runs the verification suite; the second splices those tables into this paper between marker comments. The paper is a single self-contained file with no `\input` of external resources, so it compiles anywhere as a one-file upload, and no number in it is transcribed by hand.

**`Collatz Program/Collatz_Program.py`.** The existing orbital analyser gains a *Multiverse* workspace (`multiverse_ui.py`, reached from the Multiverse button) with five panels: an interactive classification grid — Table 2 made clickable, each cell opening its own report; a per-universe report showing family, fate, reduction chain and strata; a cycle census displaying the divisor decomposition and checking the cycle equation on every cycle it finds; a symmetry explorer that places the base and transformed trajectories side by side, marking the terms the halving symmetry inserts; and heat maps of relative peak, path length, cycle count and drift.

## 11 Verification

Every theorem above is accompanied by a numerical check. Table 9 reports them. These are checks, not proofs: they establish that the statements are not false on a large finite range, and they guard the implementation against the proofs drifting apart from the code.

| check | claim | cases | result |
|---|---|---|---|
| `T-halve` | $T_{2a,2b}$ refines $T_{a,b}$: one extra even term after each odd step | 341941 | pass |
| `T-scale` | $T_{a,db}(dn) = d\,T_{a,b}(n)$ for odd $d$ | 278400 | pass |
| `T-attr` | $p \mid \gcd(a,b)$ forces every orbit into $p^{v_p(b)}\mathbb{Z}$ | 16059 | pass |
| `T-rep` | $\min(v_p(n), v_p(b))$ is an orbit invariant when $p \mid b$, $p \nmid a$ | 73632 | pass |
| `T-norm` | the normal form is coprime and idempotent | 39601 | pass |
| `T-parity` | $a+b$ odd with $a, b \geq 1$: no orbit is ever periodic | 29640 | pass |
| `T-desc` | $a = 1$, $b$ odd: every orbit cycles, least cycle element $\leq b$ | 23940 | pass |
| `T-const` | $a = 0$: exactly one cycle, of period $v_2(b) + 1$ | 199 | pass |
| `T-shape` | $b = 0$: every orbit cycles iff $a$ is a power of two, period $\log_2 a + 1$ | 39 | pass |
| `T-cyceq` | cycle equation $n_1(2^m - a^k) = b \sum_i a^{k-i} 2^{d_1+\cdots+d_{i-1}}$ | 301 | pass |
| `T-decomp` | the $d$-graded part of $\mathrm{Cyc}(T_{a,b})$ is $d \cdot \mathrm{Cyc}^*(T_{a,b/d})$ | 33 | pass |
| `T-attred` | the attracting reduction is a period-preserving bijection on cycles | 30 | pass |
| `T-tier` | knowledge ladder: OPEN = minimal with an open normal form, DECIDED = the $a = 1$ row, CLOSED = proved with closed-form cycles | 14400 | pass |
| `T-Zhalf` | $\mathbb{Z}_{>0}$ invariant iff $a + b \geq 1$; $\mathbb{Z}_{<0}$ invariant iff $b \leq a - 1$ | 3660 | pass |
| `T-Zcross` | over $\mathbb{Z}$ the sign changes only at odd $n$ with $\lvert n\rvert \leq \lvert b\rvert/a$ | 1415200 | pass |
| `T-one` | 1 is periodic iff its orbit meets a power of two | 3481 | pass |
| `T-onestep` | $a + b = 2^k$ gives a cycle through 1 of period $k + 1$ | 673 | pass |
| `T-desc2b` | every cycle of $x + b$ lies in $[1, 2b]$ | 381 | pass |
| `T-seedscale` | $n$ on a cycle of $ax + b \Rightarrow mn$ on one of $ax + mb$, for *odd* $m$ | 104 | pass |
| `T-noconj` | $cC$ is never a cycle of $T_{ca,cb}$ – the refuted conjugacy | 104 | pass |
| `T-radical` | $\mathrm{nf}(a,b) = (a,1)$ iff $\mathrm{rad}(b) \mid a$ | 7140 | pass |
| `T-qscale` | $q$-ary scaling: $F_{q\mathbf{a},q\mathbf{b}}$ refines $F_{\mathbf{a},\mathbf{b}}$ | 80808 | pass |
| `T-qtwist` | $q$-ary twisted scaling: $n \mapsto dn$ permutes the branches | 510900 | pass |
| `T-resobs` | escaping residues never divide, and diverge under the growth condition | 92510 | pass |
| `T-primmap` | primality map: every seed tested is eventually periodic, with exactly two cycles | 60000 | pass |
| `T-neg` | $T_{a,-b}(n) = -T_{a,b}(-n)$ | 9251 | pass |

Table 9: Numerical verification, run by `generate_data.py`.

## 12 The $q$-ary multiverse

Everything so far concerns $q = 2$. This section asks which of it survives when the modulus is raised, and the answer is clean: *the two symmetries survive verbatim, the parity obstruction survives in a stronger form, and the drift threshold survives with* 4 *replaced by* $q^q$. What does not survive is rigidity, and with it the uniqueness of the normal form.

Recall the map of Section 2.1: fix $q \geq 2$ and integers $a_1, \ldots, a_{q-1}$, $b_1, \ldots, b_{q-1}$, and set

$$F(n) \;=\; \begin{cases} a_i n + b_i & n \equiv i \pmod{q}, \quad 1 \leq i \leq q-1, \\ n/q & n \equiv 0 \pmod{q}. \end{cases}$$

We write $F_{\mathbf{a},\mathbf{b}}$ when the coefficients matter. This is the shape studied by Matthews and Watts [26, 27] following Hasse, and by Venturini [31] under the name *periodically linear* maps; their concern is the ergodic theory, ours the elementary reductions.

## 12.1 The two symmetries

**Proposition 12.1** ($q$-ary scaling)**.** *For every $n$,*

$$n \not\equiv 0 \pmod q \implies F_{q\mathbf{a},\,q\mathbf{b}}(n) = q \cdot F_{\mathbf{a},\mathbf{b}}(n) \text{ and } F^2_{q\mathbf{a},\,q\mathbf{b}}(n) = F_{\mathbf{a},\mathbf{b}}(n),$$

*while $F_{q\mathbf{a},q\mathbf{b}}(n) = F_{\mathbf{a},\mathbf{b}}(n) = n/q$ for $n \equiv 0$. So $F_{q\mathbf{a},q\mathbf{b}}$ traverses the same orbits as $F_{\mathbf{a},\mathbf{b}}$ with one extra dividing step inserted after each affine step.*

*Proof.* For $n \equiv i \neq 0$ the value $q(a_i n + b_i)$ is divisible by $q$, so the next step divides it back. The $n \equiv 0$ case is identical on both sides. □

**Proposition 12.2** (Twisted scaling)**.** *Let* $\gcd(d, q) = 1$ *and let* $\sigma(i) = di \bmod q$*, a permutation of* $(\mathbb{Z}/q)^\times$*-orbits on* $\{1, \ldots, q-1\}$*. Define* $a'_{\sigma(i)} = a_i$ *and* $b'_{\sigma(i)} = d\,b_i$*. Then for every* $n$*,*

$$F_{\mathbf{a}',\mathbf{b}'}(dn) \;=\; d \cdot F_{\mathbf{a},\mathbf{b}}(n).$$

*Proof.* If $n \equiv 0 \pmod q$ then $dn \equiv 0$ and both sides are $dn/q$. If $n \equiv i \neq 0$ then $dn \equiv \sigma(i) \neq 0$, because $d$ is invertible mod $q$, and $F_{\mathbf{a}',\mathbf{b}'}(dn) = a'_{\sigma(i)} dn + b'_{\sigma(i)} = d(a_i n + b_i)$. □

For $q = 2$ the only choice is $d$ odd, $\sigma$ is the identity, and Proposition 12.2 is exactly Proposition 3.2. For $q \geq 3$ something genuinely new appears: the scaling *permutes the branches*, so a universe is equivalent to one with its coefficient vector rearranged. Rigidity (Theorem 3.5) therefore fails for $q \geq 3$: two different coefficient vectors can describe the same dynamics.

**Proposition 12.3** (Attracting sublattice)**.** *Let* $m \geq 2$ *with* $\gcd(m, q) = 1$*, and suppose* $m \mid a_i$ *and* $m \mid b_i$ *for every* $i$*. Then* $m\mathbb{Z}$ *is invariant, every orbit enters it after its first affine step, and* $F$ *restricted to* $m\mathbb{Z}$ *is, via* $n \mapsto n/m$*, the map with coefficients* $(a_i, b_i/m)$*.*

*Proof.* If $m \mid n$ and $q \mid n$ then $m \mid n/q$ because $\gcd(m, q) = 1$; and $a_i n + b_i \equiv 0 \pmod m$ for every $n$ by hypothesis, which also gives the absorption after one affine step. The conjugacy is Proposition 12.2 with $d = m$ and $\sigma$ the identity on the relevant classes. □

## 12.2 The residue obstruction

For $q = 2$, "$a + b$ odd" says an odd number can never become even. The right generalisation is a finite directed graph.

**Definition 12.4.** The *residue graph* of $F$ is the map $\tau : \{1, \ldots, q-1\} \to \mathbb{Z}/q$, $\tau(i) = a_i i + b_i \bmod q$. Call $i$ *escaping* if the forward $\tau$-orbit of $i$ never reaches 0.

The graph is well defined because $n \equiv i$ forces $a_i n + b_i \equiv a_i i + b_i \pmod q$. Note it captures only the affine branches: $n/q$ is *not* determined by $n \bmod q$, since it depends on $n \bmod q^2$. That asymmetry is exactly why the obstruction is one-directional.

**Theorem 12.5** (Residue obstruction)**.** *Let $i$ be escaping and let $n \equiv i \pmod q$. Then:*

*(i) the orbit of $n$ never meets the dividing branch;*

*(ii) if in addition every residue $j$ on the forward $\tau$-orbit of $i$ satisfies $a_j \geq 2$, or $a_j = 1$ and $b_j \geq 1$, then the orbit of every $n \geq 1$ in the class of $i$ is strictly increasing, hence divergent.*

*Proof.* (i) By induction: the residue after one step is $\tau$ of the current one, and the $\tau$-orbit of $i$ avoids 0 by hypothesis, so every term is in a non-zero class and the dividing branch is never selected.

(ii) By (i) every step is affine, using the branch $j$ of the current residue. The stated condition is exactly $a_j n + b_j > n$ for all $n \geq 1$: for $a_j \geq 2$ it gives $a_j n + b_j \geq 2n > n$, and for $a_j = 1$ it gives $n + b_j > n$. So the orbit strictly increases at every step and is unbounded. □

**Corollary 12.6.** *For $q = 2$ the only affine residue is $i = 1$, $\tau(1) = a + b \bmod 2$, and $1$ is escaping iff $a + b$ is odd. Theorem 12.5 is then Theorem 6.3.*

The two conditions in Theorem 12.5 are both needed. Dropping the growth condition admits the identity branch $a_j = 1$, $b_j = 0$, whose orbits are fixed points rather than divergent, and admits $a_j = 0$, whose branch is constant. A machine check over $q \leq 5$ and coefficients in $\{0, \ldots, 3\}$ found 239,530 orbits from escaping residues, none of which ever divided, confirming (i); and 125,765 orbits satisfying the growth condition, all strictly increasing, confirming (ii).

### 12.3 Drift

**Heuristic 12.7.** Suppose the residues equidistribute along orbits and that an affine step whose image is divisible by $q$ is followed by $q/(q-1)$ divisions on average. Then the geometric drift per affine step is

$$\delta \;=\; \frac{\left(\prod_{i=1}^{q-1} a_i\right)^{1/(q-1)}}{q^{\,q/(q-1)}},$$

so $F$ contracts on average precisely when

$$\prod_{i=1}^{q-1} a_i \;<\; q^{\,q}.$$

For $q = 2$ this is $a < 4$, the threshold of Section 9, and it recovers $\delta = a/4$. For $q = 3$ the condition is $a_1 a_2 < 27$: the map with $a = (3, 3)$ has $\delta \approx 0.577$ and should collapse, while $a = (5, 7)$ has $\delta \approx 1.139$ and should escape. As at $q = 2$ this is a heuristic and nothing more; the equidistribution it assumes is not known for any single map.

**Remark 12.8.** The threshold $q^q$ grows so fast that the "interesting" region — where the drift is near 1 — is a vanishing fraction of coefficient space as $q$ grows, but it is never empty: taking $a_i \approx q^{q/(q-1)}$ for all $i$ puts $\delta$ near 1. The $q$-ary multiverse therefore has a Goldilocks region at every modulus, and $3x + 1$ is merely its smallest member.

## 13 Beyond modularity

Modularity is doing an enormous amount of work in everything above. This section asks what happens without it. The answer is that essentially nothing survives, but the reason is interesting and it suggests a principle we state as a conjecture.

**Definition 13.1** (Predicate map)**.** Let $P_1, \ldots, P_k$ be predicates on $\mathbb{Z}$ with $P_k$ always true, and let $f_1, \ldots, f_k$ be functions $\mathbb{Z} \to \mathbb{Z}$. The associated *predicate map* sends $n$ to $f_j(n)$ for the least $j$ with $P_j(n)$.

Every map $\mathbb{Z} \to \mathbb{Z}$ is a predicate map, so there is no classification theorem to be had; and by Conway [9] even the affine, congruence-driven case is universal once enough branches are allowed. The question worth asking is therefore not "what are the predicate maps like" but *which predicates preserve the theory.*

**Proposition 13.2** (Congruence predicates change nothing)**.** *If every $P_j$ is a union of residue classes modulo some $M_j$, and every $f_j$ is affine, then the predicate map is a periodically linear map with modulus $M = \operatorname{lcm}(M_1, \ldots, M_k)$, and if one branch is $n \mapsto n/q$ on the class $0 \bmod q$ with $q \mid M$ it is a $q$-ary map after refining the classes. All of Section 12 applies.*

*Proof.* Each $P_j$ is determined by $n \bmod M$, so the branch taken is a function of $n \bmod M$; grouping the classes gives the stated shape. □

So the theory covers exactly the *eventually periodic* predicates. The moment a predicate is not a congruence condition — primality, squarefreeness, being a perfect power, a digit condition in a base coprime to the branch moduli — every tool above fails at once: the residue graph does not exist, Terras-style class refinement has nothing to refine, and the affine bookkeeping that gives the cycle equation has no periodic structure to lean on.

### 13.1 A worked example: the primality map

The programme that produced this paper experimented with

$$f(n) \;=\; \begin{cases} 3n+1 & n \text{ prime}, \\ n/2 & n \text{ even}, \\ n+1 & \text{otherwise}, \end{cases}$$

and observed that every orbit it tried was eventually periodic. We can say rather more.

**Proposition 13.3.** *For the map above: every odd prime $p$ satisfies $f^2(p) = (3p+1)/2 \approx 1.5\,p$, and every odd composite $m$ satisfies $f^2(m) = (m+1)/2 \approx 0.5\,m$. Hence an orbit can diverge only if the proportion $r$ of* prime *values among the odd numbers it visits satisfies*

$$\left(\tfrac{3}{2}\right)^r \left(\tfrac{1}{2}\right)^{1-r} > 1, \qquad \textit{i.e.} \qquad r \;>\; \frac{\ln 2}{\ln 3} \;=\; 0.63092\ldots$$

*Proof.* An odd $n$ is sent to $3n+1$ or $n+1$, both even, and the next step halves. Taking logarithms of the product of the per-odd-step multipliers gives the stated condition. ☐

A search over all seeds $n \le 200{,}000$ finds that *every* orbit is eventually periodic, and that there are exactly two cycles:

$C_1 = \{2, 4, 5, 7, 8, 10, 11, 13, 16, 17, 20, 22, 26, 34, 40, 52\}$, period 16, 86.9% of seeds,
$C_2 = \{31, 41, 47, 62, 71, 81, 82, 94, 107, 124, 142, 161, 162, 214, 322\}$, period 15, 13.1%.

The tempting argument — "primes have density 0, so the growing branch is negligible" — is too quick, and measuring carelessly makes it look false. Counting prime values over whole orbits gives densities of 55–71%, alarmingly close to the threshold, because the two cycles are short and prime-rich and orbits spend most of their time inside them. Restricting to the transient part and to values above a floor gives the honest picture:

| values above | odd values seen | prime | density |
|---:|---:|---:|---:|
| $10^3$ | 3958 | 1328 | 33.6% |
| $10^4$ | 3500 | 723 | 20.7% |
| $10^6$ | 2481 | 344 | 13.9% |
| $10^8$ | 3177 | 331 | 10.4% |
| $10^{11}$ | 1270 | 107 | 8.4% |

The density decays, but it sits consistently around 2–3 times the $1/\ln n$ one would predict for random integers of that size (3.6% at $10^{12}$) — the map's odd values are biased towards primes, because $(m+1)/2$ and $(3p+1)/2$ avoid small factors more often than a random integer does. The bias is real and worth noting; it is also nowhere near enough. At $10^{12}$ the observed 8.4% must reach 63.1% for divergence, and it is falling.

**Conjecture 13.4** (Primality map)**.** Every orbit of $f$ is eventually periodic, and the two cycles listed above are the only ones.

### 13.2 The density–drift principle

The computation above is an instance of something general, which we state as the organising conjecture of this section.

**Conjecture 13.5** (Density–drift principle)**.** Let $T$ be a predicate map whose branches are asymptotically multiplicative, $f_j(n) \sim c_j n$, and suppose that along orbits the branch $j$ is taken with asymptotic frequency $d_j$, $\sum_j d_j = 1$. Then almost every orbit is bounded if $\prod_j c_j^{d_j} < 1$ and almost every orbit diverges if $\prod_j c_j^{d_j} > 1$.

For congruence predicates the frequencies $d_j$ are the densities of the classes and the principle specialises to Heuristic 12.7 and, at $q = 2$, to the classical $a/4$. Its content in the non-modular case is the assertion that the branch frequencies along orbits exist at all — which for the primality map is the statement that the orbit does not conspire to visit primes more often than a density argument allows, and which the table above supports without settling.

**Conjecture 13.6** (Sparse growth)**.** If the branches with $c_j > 1$ are selected by predicates whose density along orbits tends to 0, and some branch with $c_j < 1$ is taken with frequency bounded below, then every orbit is bounded.

Conjecture 13.6 would give Conjecture 13.4 immediately, since the primes have density 0. We state it separately because it is the weakest statement we can see that does the job, and because it isolates the difficulty: it is a statement about the *invariant measure* of the map, not about the density of primes in $\mathbb{N}$.

**Conjecture 13.7** (Reduction boundary)**.** A predicate map admits a nontrivial reduction theory — symmetries of the kind in Section 3, a normal form, and a finite obstruction graph — if and only if its predicates are eventually periodic.

The forward direction is Proposition 13.2. The converse is the substantial half, and Conway's universality [9] together with the $\Pi^0_2$-completeness of Kurtz and Simon [10] is evidence for it: a family rich enough to encode arbitrary computation cannot admit a classification, and predicate maps are certainly that rich. What would make Conjecture 13.7 a theorem is a precise definition of "reduction theory", which we do not have.

**Remark 13.8.** This section is deliberately short. The point we want to record is structural: the reason $3x + 1$ is attackable at all — the reason there is a Terras argument, a cycle equation, a normal form — is that its branch predicate is a congruence. Change that, and the problem does not become harder in degree; it changes kind, from number theory to computability.

## 14 Conjectures, and which of them survive

The programme that produced this paper accumulated a list of conjectures. It is worth publishing the list together with its casualties: three of them are false, and the counterexamples are small enough to check by hand.

- **C1** *(Generalised Crandall conjecture.)* For every $b \geq 1$ and every odd $a > 3$, the map $T_{a,b}$ has an orbit that never reaches a cycle through 1. **Open.** For $b = 1$ this is Crandall's conjecture [4], known for almost all $a$ by Franco and Pomerance [5] and for $a = 2^k - 1$ by Santos [6]. By Theorem 5.3 it suffices to consider $(a, b)$ in normal form, and by Corollary 8.3 the cycle structure for composite $b$ is determined by that of its divisors — so the conjecture really only concerns primitive universes.

**C2** *Every orbit of every map with $a < 3$ is bounded.* **False as stated, and completely settled.** It fails for $a = 2$, $b$ odd — $2x + 1$ sends $1 \to 3 \to 7 \to 15 \to \cdots$ — which is Theorem 6.3. Restricted to the cases where it is not refuted by parity it is a theorem: $a = 0$ is Proposition 6.1, $a = 1$ with $b$ odd is Theorem 6.4, and $a = 2$ with $b$ even reduces to $a = 1$ by Proposition 3.1.

**C3** *The q-ary map of Section 2.1 is not Turing-complete.* **False for the unrestricted family.** Conway [9] constructed universal generalized Collatz functions, and Kurtz and Simon [10] showed the natural reachability problem is $\Pi^0_2$-complete. We note honestly that Conway's construction uses rational multipliers chosen so that the image is an integer, which is more freedom than the shape in Section 2.1 allows; whether that exact restricted shape is universal we do not know, and we regard **C3** as open only in that narrow reading.

**C4** *If* 1 *lies on a cycle $C$ containing another odd number, and no scaling relation applies, then no odd number outside $C$ enters $C$.* **False.** In $5x+1$ the cycle through 1 is $\{1, 6, 3, 16, 8, 4, 2\}$, which contains the odd number 3; the offset $b = 1$ admits no scaling; and the odd number $15 \notin C$ enters it:

$$15 \to 76 \to 38 \to 19 \to 96 \to 48 \to 24 \to 12 \to 6 \in C.$$

A search over odd $a \leq 39$, odd $b \leq 39$ and odd seeds below 600 finds eight such violations, the smallest in the most-studied universe of the whole family.

**C5** 1 *lies on a cycle of every map* $3x + b$. **False**, as the programme already recorded: in $3x + 3$ the orbit of 1 is $1 \to 6 \to 3 \to 12 \to 6$, which cycles without returning to 1.

**C5′** 1 *lies on a cycle of* $3x + b$ *whenever* $b$ *is not a power of* 3. **False.** The smallest counterexample is $b = 7$: the orbit of 1 is $1 \to 10 \to 5 \to 22 \to 11 \to 40 \to 20 \to 10$, which enters the cycle $\{10, 5, 22, 11, 40, 20\}$ and never returns to 1. Further counterexamples are $b = 15, 19, 21, 23, 25, 31, 33, 35, 37, 39, 45, \ldots$ Theorem 6.7 replaces the conjecture with an exact criterion: 1 lies on a cycle of $3x + b$ if and only if the orbit of 1 meets a power of two, and $3 + b = 2^k$ (Corollary 6.8) is the one-step case that accounts for 93 of the 147 instances we found.

**Q1** *Does* $3511x + 1$ *have an orbit avoiding* 1*?* 3511 is the second known Wieferich prime; Crandall's conjecture is known for the first, 1093 [4]. **Open**, and outside the reach of anything here: our reductions never change the multiplier (Theorem 3.5), so the row $a = 3511$ is untouched by them.

**Remark 14.1.** The pattern in **C4** and **C5′** is worth naming. Both conjectures generalise from the universes where a scaling relation is doing the work, and both break exactly where it stops. This is the practical value of Theorem 5.2: once a universe is in normal form, one knows that no further structure is hiding, and a pattern that persists there is a pattern about the universe rather than about its presentation.

## 15 What is *not* proved

1. **No primitive universe is resolved.** Theorem 6.10 case (4) is a statement of ignorance. Nothing here bears on whether $3x + 1$ has other cycles, whether any $3x + b$ orbit diverges, or whether $5x + 1$ has a divergent orbit. The reductions move problems around; they do not solve any.

2. **Cycle counts are lower bounds.** Every count in Section 8 comes from a search over a finite seed window with a finite step budget. A cycle whose basin misses the window is not

found. We report the window with every table and we checked stability by widening it, but stability under widening is not a proof of completeness. Artem makes the same caveat; it is a real one, and $b = 71$ in Table 6 is a concrete example of a count that changes when the window grows.

3. **The drifts are heuristics.** $\delta_{\text{odd}} = a/4$ rests on $\mathbb{E}[v_2(an + b)] = 2$, which assumes an equidistribution that is not proved for any particular universe. It predicts nothing about any individual orbit. Tao's result [41] is the state of the art on making such heuristics rigorous for $3x + 1$, and it falls well short of the conjecture.

4. **Densities are natural densities.** Theorem 7.4 says nothing about any particular $(a, b)$, and in particular does not say that a positive proportion of universes will remain open — only that a positive proportion is open *now*.

5. **The $q$-ary chapter is elementary.** Section 12 proves only the two symmetries and the residue obstruction. It does not construct a normal form for $q \geq 3$, and it cannot: rigidity fails there, because twisted scaling permutes the branches (Proposition 12.2), so a universe has several equally valid coefficient vectors. Building the analogue of Theorem 5.2 modulo that permutation action is the obvious next problem, and we have not solved it.

6. **Section 13 proves almost nothing.** Its content is one proposition, one computation, and three conjectures. In particular Conjecture 13.5 is not even precisely stated — "the branch frequency along orbits" is not known to exist — and Conjecture 13.7 awaits a definition of "reduction theory". We include the chapter because the boundary it points at seems to us the right one, not because we can defend it.

7. **Only affine equivalences are classified.** Theorem 3.5 rules out affine intertwinings. It does not rule out more exotic conjugacies, and it does not rule out that two rows are related by an argument that is not a change of variable at all.

8. **$b < 0$ is only transported, not studied.** Proposition 3.4 reduces negative offsets to the negative integers of a positive-offset universe, but we do not classify the dynamics there; $3x - 1$ has at least three cycles and is as open as $3x + 1$.

# 16 Attribution

The multiverse framing and a large fraction of the phenomena in this paper were found, experimentally and independently of the literature, by carykh [1, 2] and by Artem [3]. This paper's contribution is to prove them, to unify them, and to draw the line between solved and open. Table 10 attributes each result.

**Remark 16.1** (Corrections to the earlier notes)**.** Five items needed amending, and all five are recorded above and in §14.

1. The $b = 0$ cycle length is $\log_2(a) + 1$, not $\log_2(a) + 2$ (Proposition 6.2); the discrepancy is the convention of listing the starting term twice.

2. The $a = 1$ family was recorded as having "two loops". The two named always exist, but there are frequently more, first at $b = 7$ (Remark 6.6).

3. The cycle-seed scaling $n \mapsto mn$ requires $m$ *odd.* For even $m$ the image $mn$ is even and simply halves; a search over $a \in \{3, 5, 7\}$, $b \leq 7$ found 140 failures for even $m$ and none for odd $m$.

4. The claim that a $v_2$-preserving $g$ conjugates $T$ to $T_g$ is false (Proposition 3.8); 0 of 104 test cases lift.

| observation | first stated | status here |
|---|---|---|
| The multiverse framing; the $(a, b)$ grid | carykh [1] | adopted |
| Hub-and-spoke family, $a = 0$ | carykh | Prop. 6.1: proof, exact period $v_2(b) + 1$, uniqueness |
| Shape family $b = 0$ with $\log_2(a) + 1$ sides | carykh | Prop. 6.2: proof, and the $u \geq 3$ runaway |
| Runaway checkerboard, $a + b$ odd; linear vs exponential subtypes | carykh | Thm. 6.3: proof |
| Even parallel universes $(2a, 2b) \sim (a, b)$ | carykh | Prop. 3.1: exact orbit refinement, period formula, cycle bijection |
| Odd parallel universes, offset scaled by odd $d$ | carykh | Prop. 3.2: proof |
| Connected vs disconnected odd parallels | carykh (named), Artem (criterion) | Thm. 4.1, Cor. 4.2: the criterion is $p \mid a$, per odd prime, with absorption time $v_p(b)$ |
| "Factor loyalty" on the odd diagonal $a = b$ | carykh | special case of Thm. 4.1(i) |
| $3x + 9$ attracts in two odd steps; $3x + 3^j$ in $j$ | Artem | Thm. 4.1(i) |
| $3x + 15$ is a copy of $3x + 5$, not of $3x + 1$ | Artem | Thm. 5.2, Thm. 5.3 |
| Composite offsets contain a copy of every divisor | Artem | Thm. 8.2, Cor. 8.3: strengthened to an exact partition of the phase space and of the cycle set |
| Cycle counts for $3x + b$, $b < 100$ | Artem | Table 6: recomputed at a converged window |
| Only powers of 3 away from a base have one cycle | Artem | Cor. 8.6 |
| No $b < 100$ gives exactly seven cycles | Artem | confirmed at a wider window (§8) |
| Drift $\sqrt{a}/2$; Goldilocks / Icarus / black-hole rows | carykh | §9, with the equivalent form $a/4$ per odd step and the threshold $a < 4$ |
| $(k, m) = (3, 5)$ and $(17, 27)$ in $3x + 5$, from $3^k \approx 2^m$ | carykh | Thm. 8.1 |
| $5x + 1$ and $5x + 5$ share the fingerprint $7, 10, 10$ | carykh | Prop. 3.2 + Thm. 4.1(i) |
| Average max growth heat map; $\sqrt{2}$ on the $a = 2$ row | carykh | Table 8 |
| The $6x + 34$ anomaly is a threshold artefact | carykh | confirmed, §9 |
| Nine "unique" universes in the $6 \times 6$ grid | carykh | Thm. 7.1: three solved, two reducible, four open |
| *From the authors' own earlier programme (research notes and the repository's `proofs.txt`):* | | |
| Hub-and-spoke $(a = 0)$, shapes $(b = 0)$, parity runaways, and the $a = 1$ row | the authors | Props. 6.1, 6.2, Thms. 6.3, 6.4: proofs, plus two corrections noted below |
| $n$ seeds a cycle of $ax + b \Rightarrow mn$ seeds one of $ax + mb$ | the authors | Prop. 3.2; **needs $m$ odd** — see below |
| $a = 2^k - b$ gives a cycle through 1 | the authors | Cor. 6.8, and strengthened to an exact criterion in Thm. 6.7 |
| Every cycle of $x + b$ lies in $[1, 2b]$ | the authors | Thm. 6.4(iii); this is what makes Table 1 exact |
| $2ax + 2b$ is equivalent to $ax + b$ | the authors, and carykh independently | Prop. 3.1 |
| $n$ in $ax + b$ corresponds to $n/b$ in $ax + 1$; equivalence iff $b \mid a$ | the authors | Prop. 3.2, Prop. 5.5; the correct criterion for equivalence is $\mathrm{rad}(b) \mid a$ (Thm. 5.6) |
| Post-composing the odd branch: $T_g(n) = g(T(n))$ for odd $n$ | the authors | Prop. 3.7 |
| $g$ conjugates $T$ to $T_g$ when $v_2$ is preserved | the authors | **refuted**, Prop. 3.8 |
| The conjecture list C1–C5′, Q1 | the authors | §14: C2, C4 and C5′ refuted, C3 refuted in the unrestricted reading |
| The $q$-ary framing (mod $q$ rather than mod 2), and the programme of pushing the theory beyond modularity | the authors | §12, §13; the $q$-ary shape itself is Matthews–Watts [26] |
| The primality-driven map $3n + 1$ / $n/2$ / $n + 1$ | the authors | Prop. 13.3, Conj. 13.4: the threshold $\ln 2/\ln 3$ and the measured densities |
| Rigidity of affine equivalences | — | Thm. 3.5 |
| Normal form; termination and confluence | — | Thm. 5.2, Thm. 5.3 |
| Master classification | — | Thm. 6.10 |
| Densities $1/3$ and $2/\pi^2$ | — | Thm. 7.4 |
| $q$-ary scaling, twisted scaling, residue obstruction, threshold $q^q$ | — | Props. 12.1, 12.2, Thm. 12.5, Heur. 12.7 |
| The density–drift principle and the reduction boundary | — | Conjs. 13.5, 13.6, 13.7 |
| $a = 1$ solved with the bound $\min C \leq b$ | — | Thm. 6.4 |

5. Conjectures C2, C4 and C5′ are false, with counterexamples $2x + 1$, $5x + 1$ and $3x + 7$ respectively (§14).

None of these affects the reductions or the classification.

**Remark 16.2.** Several of the results here are certainly known in some form to specialists. The cycle equation (Theorem 8.1) is classical, appearing in Lagarias [38] for $3x + 1$. The scaling relation (Proposition 3.2) is folklore. What we have not found stated anywhere is the combination: the normal form, the resulting master classification, and the two densities.

# 17 Conclusion

The generalised Collatz multiverse is more structured than it looks and smaller than it looks. Two symmetries — halving and odd scaling — and one dichotomy — whether an odd prime divides the multiplier or only the offset — reduce every universe to a coprime normal form. Four of the five resulting cases are solved completely and elementarily. The fifth, $a \geq 3$ odd with $b$ odd and coprime, is open, has density $2/\pi^2$, and contains $3x + 1$.

Pushing outward sharpens rather than dilutes that picture. Raising the modulus keeps both symmetries and replaces the parity obstruction by a finite directed graph on $\mathbb{Z}/q$, but costs rigidity: scaling permutes the branches, so there is no normal form at $q \geq 3$ and we do not know what should replace it. Dropping modularity altogether costs everything. The reduction theory, the cycle equation and the Terras-style refinement all rest on the branch being selected by a congruence, and once it is not — primality, say — the family is rich enough to encode arbitrary computation, so no classification can exist. On that reading the boundary of the subject is not a difficulty of degree but a change of kind, and it runs exactly along periodicity of the branch predicate.

The videos that prompted this paper end with carykh's speculation that Collatz chose $3x + 1$ "not because that one was particularly interesting, but because it was the only rule set that wasn't obvious in some way". The correct version is narrower and, we think, more striking: the drift threshold $a < 4$ admits exactly two odd multipliers, one of which is solved by a two-line descent argument. $3x + 1$ is not the only interesting universe, but it is the smallest member of the only row where the difficulty is of the Collatz kind.

# Note on the use of artificial intelligence

The research programme, the earlier hand computations, and the existing software in the repository are the author's. The theorems and proofs in this paper, the library `multiverse.py`, the verification suite, the Multiverse workspace in `Collatz_Program.py`, and the text of this paper were produced in collaboration with an artificial-intelligence system (Claude, Anthropic), at the author's direction.

The proofs here are hand-checkable and short by design, and each is accompanied by an independent numerical check reported in Table 9; the cycle censuses were cross-checked against the counts published by Artem [3] and carykh [1, 2], which agree in every case where the search windows overlap. Section 15 states explicitly what is not proved. Unlike the companion paper [42] in this repository, this development is *not* machine-checked in a proof assistant; the reader should hold its theorems to the ordinary standard of a hand-written proof.

# A The ccchallenge corpus, classified

ccchallenge.org [19] is a collaborative project to formalise the Collatz literature in proof assistants; it maintains a catalogue with a public API. We took a snapshot of 375 papers and classified them by keyword into the domains below, first match winning, so that each paper is counted once. The classification is produced by `Multiverse/literature.py` from the cached snapshot `data/ccchallenge_papers.json`, and the ordering of the domain list is significant: generalised maps are tested first, because those are the papers this paper engages with.

Table 11 gives the resulting counts.

| domain | papers |
|---|---|
| generalised maps | 49 |
| cycles and lower bounds | 32 |
| stopping time and density | 15 |
| computation and verification | 53 |
| logic, automata, formalisation | 8 |
| p-adic, ergodic, analytic | 13 |
| surveys and bibliographies | 8 |
| other $3x + 1$ | 197 |
| **total** | **375** |

Table 11: The corpus by domain.

**A caution.** The corpus is a catalogue, not a curated bibliography: it includes preprints and self-published items alongside refereed work. Two entries claim a complete proof of the conjecture in a venue with no evidence of refereeing (Halemane, 2025; Novgorodtsev, 2026); neither is formalised on ccchallenge, and we list them for completeness without endorsement. A third, Bruckman (2008), appeared in a refereed venue and its argument is generally regarded as flawed. Nothing in this paper depends on any of them. Readers wanting a curated entry point should use Lagarias' annotated bibliographies [15, 16, 17].

## A.1 The generalised-map class

These are the 49 papers our classifier assigns to generalised maps — the part of the corpus relevant to the multiverse. Those we engage with individually are cited in Section 1.2.

| author (year) | title |
|---|---|
| *Heppner* (1978) | Eine Bemerkung zum Hasse-Syracuse Algorithmus |
| *Steiner* (1981) | On the "$Qx + 1$" Problem, $Q$ odd |
| *Steiner* (1981) | On the "$Qx + 1$" Problem, $Q$ odd II |
| *Matthews & Watts* (1984) | A generalization of Hasse's generalization of the Syracuse algorithm |
| *Lagarias* (1985) | The $3x + 1$ problem and its generalizations |
| *Matthews & Watts* (1985) | A Markov approach to the generalized Syracuse algorithm |
| *Leigh* (1986) | A Markov process underlying the generalized Syracuse algorithm |
| *Dolan et al.* (1987) | A generalization of Everett's result on the Collatz $3x + 1$ problem |
| *Matthews & Leigh* (1987) | A generalization of the Syracuse algorithm to $F_q[x]$ |
| *Buttsworth & Matthews* (1990) | On some Markov matrices arising from the generalized Collatz mapping |
| *Franco* (1990) | Diophantine Approximation and the $qx + 1$ Problem |
| *Zhang & Yu* (1990) | A generalization of Kakutani's conjecture |
| *Venturini* (1992) | Iterates of Number Theoretic Functions with Periodic Rational Coefficients (Generalization of the $3x + 1$ problem) |
| *Korec* (1992) | The $3x + 1$ Problem, Generalized Pascal Triangles, and Cellular Automata |
| *Matthews* (1992) | Some Borel measures associated with the generalized Collatz mapping |
| *Wirsching* (1994) | A Markov chain underlying the backward Syracuse algorithm |
| *Mignosi* (1995) | On a Generalization of the $3x + 1$ Problem |
| *Brocco* (1995) | A Note on Mignosi's Generalization of the $3x + 1$ Problem |

| author (year) | title |
|---|---|
| *Franco & Pomerance* (1995) | On a Conjecture of Crandall Concerning the $qx+1$ Problem |
| *Bernstein & Lagarias* (1996) | The $3x+1$ Conjugacy Map |
| *Venturini* (1997) | On a Generalization of the $3x+1$ problem |
| *Belaga & Mignotte* (1999) | Embedding the $3x+1$ Conjecture in a $3x+d$ Context |
| *Garcia & Tal* (1999) | A note on the generalized $3n+1$ problem |
| *Belaga & Mignotte* (2000) | Cyclic Structure of Dynamical Systems Associated with $3x+d$ Extensions of Collatz Problem |
| *Dumont & Reiter* (2001) | Visualizing Generalized 3x+1 Function Dynamics |
| *Ke & Wei* (2001) | Generalization Concerning the $3x+1$ Problem |
| *Zarnowski & E.* (2001) | Generalized inverses and the total stopping time of Collatz sequences |
| *Rouet & Feix* (2002) | A generalization of the Collatz problem. Building cycles and a stochastic approach |
| *Belaga* (2003) | Effective polynomial upper bounds to perigees and numbers of $(3x+d)$-cycles of a given oddlength |
| *Wang et al.* (2003) | The distribution of the fixed points on the real axis of a generalized $3x+1$ function and some related fractal images |
| *Michel & Pascal* (2004) | Small Turing machines and generalized busy beaver competition |
| *Ohira & Yamashita* (2004) | A Generalization of the Collatz problem |
| *Farkas* (2005) | Variants of the $3N+1$ problem and multiplicative semigroups |
| *Matthews & R.* (2005) | The generalized $3x+1$ mapping |
| *Belaga & Mignotte* (2006) | The Collatz problem and its generalizations: Experimental Data. Table 1. Primitive cycles of $3x+d$ mappings |
| *Belaga & Mignotte* (2006) | The Collatz problem and its generalizations: Experimental Data. Table 2. Factorization of Collatz numbers $2^l-3^k$ |
| *Simons* (2007) | A simple (inductive) proof for the non-existence of 2-cycles for the $3x+1$ problem |
| *Kurtz & Simon* (2007) | The undecidability of the generalized Collatz problem |
| *Wang & Yu* (2007) | Visualizing generalized $3x+1$ function dynamics based on fractal |
| *Snapp & Tracy* (2008) | The Collatz problem and Analogues |
| *Simons* (2008) | On the (non)-existence of $m$-cycles for generalized Syracuse sequences |
| *Hicks et al.* (2008) | A Polynomial Analogue of the $3N+1$ Problem? |
| *Wang et al.* (2008) | Dynamics of the generalized $3x+1$ function determined by its fractal images |
| *Muller & Helmut* (2009) | Uber Periodenlangen and die Vermutungen von Collatz und Crandall |
| *Capco* (2015) | Odd Collatz Sequence and Binary Representations |
| *Santos* (2020) | On the Collatz general problem qn+1 |
| *Yolcu et al.* (2021) | An automated approach to the Collatz conjecture. |
| *Siegel* (2024) | p, q-adic Analysis and the Collatz Conjecture |
| *Novgorodtsev* (2026) | Algebraic Proof of the Collatz Conjecture and Universal Classification of 2-Adic Dynamical Systems |

## A.2 The full snapshot

For completeness, the whole corpus, tagged by domain: **G** generalised maps, **C** cycles and lower bounds, **D** stopping time and density, **V** computation and verification, **L** logic, automata, formalisation, **P** $p$-adic, ergodic, analytic, **S** surveys, – other $3x+1$.

| | author | year | title |
|---|---|---|---|
| - | Slakmon & Macot | 2006 | On the almost convergence of Syracuse sequences |
| G | Novgorodtsev | 2026 | Algebraic Proof of the Collatz Conjecture and Universal Classifica ... |
| D | Novgorodtsev | 2026 | Exact Synchronized Coalescence Clocks and Peak-Envelope Transfer f ... |
| - | Kontorovich & Miller | 2005 | Benford's law, values of $L$-functions, and the $3x+1$ problem |
| - | Kontorovich & Sinai | 2002 | Structure Theorem for $(d,g,h)$-maps |
| V | Stokes | 2021 | The search for self-contained numbers: k-special 3-smooth represen ... |
| C | Amleh et al. | 1998 | On some difference equations with eventually periodic solutions |
| - | Atkin | 1966 | Comment on Problem $63-13^*$ |

| | author | year | title |
|---|---|---|---|
| - | Baker & A. | 1966 | Linear forms in the logarithms of algebraic numbers |
| - | Brent | 2002 | $3X + 1$ dynamics on rationals with fixed denominator |
| G | Snapp & Tracy | 2008 | The Collatz problem and Analogues |
| C | Seifert | 1988 | On the arithmetic of cycles for the Collatz-Hasse ('Syracuse') con ... |
| P | Kraft & Monks | 2010 | On conjugacies for the $3x + 1$ map induced by continuous endomorphi ... |
| - | Wiggin | 1988 | Wondrous Numbers – Conjecture about the $3n + 1$ family |
| - | Hong | 1986 | About $3X + 1$ problem |
| V | Hayes | 1984 | Computer recreations: The ups and downs of hailstone numbers |
| - | Thwaites | 1985 | My conjecture |
| - | Thwaites | 1996 | Two Conjectures, or how to win *pounds* 1000 |
| C | Chisala | 1994 | Cycles in Collatz Sequences |
| - | Everett | 1977 | Iteration of the number theoretic function $f(2n) = n, f(2n + 1) = 3n + 2$ |
| S | Ogilvy | 1972 | Tomorrow's Math: unsolved problems for the amateur |
| - | Viola | 1983 | Un Problema di Aritmetica (A problem of arithmetic) |
| - | Inclan | 2026 | Analysis of Collatz Trajectories via Modulo 6 Modular Families and ... |
| - | Inclan | 2026 | Inverse Collatz map study, version 3 |
| C | Farruggia et al. | 1996 | The elimination of a family of periodic parity vectors in the $3x+$ ... |
| V | Cadogan | 1996 | Exploring the $3x + 1$ problem I |
| - | Ashbacher | 1992 | Further Investigations of the Wondrous Numbers |
| - | Cadogan | 1984 | A note on the $3x + 1$ problem |
| V | Cadogan | 1991 | Some observations on the $3x + 1$ problem |
| V | Cadogan | 2000 | The $3x + 1$ problem: towards a solution |
| V | Cadogan | 2003 | Trajectories in the $3x + 1$ problem |
| - | Cadogan | 2006 | A Solution to the $3x + 1$ Problem |
| - | Trigg et al. | 1976 | Comments on Problem 133 |
| C | Hercher | 2023 | There are no Collatz $m$-Cycles with $mleq91$ |
| - | Baiocchi | 1998 | 3N+1, UTM e Tag-Systems |
| - | Pickover | 1989 | Hailstone $3n + 1$ Number graphs |
| C | Bohm & Sontacchi | 1978 | On the existence of cycles of given length in integer sequences li ... |
| - | Levy | 2004 | Injectivity and Surjectivity of Collatz Functions |
| - | Rawsthorne | 1985 | Imitation of an iteration |
| P | Bernstein | 1994 | A Non-Iterative 2-adic Statement of the $3x + 1$ Conjecture |
| G | Bernstein & Lagarias | 1996 | The $3x + 1$ Conjugacy Map |
| - | Shanks | 1965 | Comments on Problem $63 - 13$* |
| - | Shanks | 1975 | Problem #5 |
| - | Merlini & Sala | 1999 | On the Fibonacci's Attractor and the Long Orbits in the $3n + 1$ Pro ... |
| D | Klarner | 1981 | An algorithm to determine when certain sets have 0 density |
| - | Klarner | 1982 | A sufficient condition for certain semigroups to be free |
| - | Klarner | 1988 | $m$-recognizability of sets closed under certain affine functions |
| - | Klarner & Post | 1992 | Some fascinating integer sequences |
| - | Klarner & Rado | 1973 | Linear combinations of sets of consecutive integers |
| - | Klarner & Rado | 1974 | Arithmetic properties of certain recursively defined sets |
| D | Applegate & Lagarias | 1995 | Density Bounds for the $3x + 1$ Problem I. Tree-Search Method |
| D | Applegate & Lagarias | 1995 | Density Bounds for the $3x + 1$ Problem II. Krasikov Inequalities |
| V | Applegate & Lagarias | 1995 | The distribution of $3x + 1$ trees |
| D | Applegate & Lagarias | 2003 | Lower bounds for the total stopping time of $3x + 1$ iterates |
| - | Applegate & Lagarias | 2006 | The $3x + 1$ semigroup |
| V | Barina | 2025 | Improved verification limit for the convergence of the Collatz con ... |
| - | Gale | 1991 | Mathematical Entertainments: More Mysteries |
| - | Gluck & Taylor | 2002 | A new statistic for the $3x + 1$ problem |
| V | Kay | 1972 | An Algorithm for Reducing the Size of an Integer |
| - | Boyd | 1985 | Which rationals are ratios of Pisot sequences? |
| - | Clark | 1995 | Second-Order Difference Equations Related to the Collatz $3n + 1$ Co ... |
| C | Clark | 2006 | Periodic solutions of arbitrary length in a simple integer iteration |
| V | Clark & Lewis | 1995 | A Collatz-Type Difference Equation |
| V | Clark & Lewis | 1998 | Symmetric solutions to a Collatz-like system of Difference Equati ... |
| - | Hoffman & Klarner | 1978 | Sets of integers closed under affine operators- the closure of fin ... |
| - | Hoffman & Klarner | 1979 | Sets of integers closed under affine operators- the finite basis t ... |
| - | Domenici | 2009 | A few observations on the Collatz problem |
| - | Coppersmith | 1975 | The complement of certain recursively defined sets |
| - | Shaw | 2006 | The pure numbers generated by the Collatz sequence |
| - | Marcu | 1991 | The powers of two and three |
| - | Belaga | 1998 | Reflecting on the $3x + 1$ Mystery: Outline of a Scenario- Improbabl ... |
| G | Belaga | 2003 | Effective polynomial upper bounds to perigees and numbers of $(3x+$ ... |
| G | Belaga & Mignotte | 1999 | Embedding the $3x + 1$ Conjecture in a $3x + d$ Context |
| G | Belaga & Mignotte | 2000 | Cyclic Structure of Dynamical Systems Associated with $3x + d$ Exten ... |
| V | Belaga & Mignotte | 2006 | Walking Cautiously into the Collatz Wilderness: Algorithmically, N ... |

| | author | year | title |
|---|---|---|---|
| G | Belaga & Mignotte | 2006 | The Collatz problem and its generalizations: Experimental Data. Ta ... |
| G | Belaga & Mignotte | 2006 | The Collatz problem and its generalizations: Experimental Data. Ta ... |
| L | Lehtonen | 2008 | Two undecidable versions of Collatz's problems |
| - | Eliahou et al. | 2022 | Is the Syracuse Falling Time Bounded by 12? |
| P | Jesus | 2026 | A Global Occupation Conjecture for the Accelerated 3x + 1 Map with ... |
| - | Oliveri & Vella | 1998 | Alcune Questioni Correlate al Problema Del "3D+1" |
| G | Yolcu et al. | 2021 | An automated approach to the Collatz conjecture. |
| - | Barone | 1999 | A heuristic probabilistic argument for the Collatz sequence |
| - | Roosendaal | 2004 | On the $3x + 1$ problem |
| G | Heppner | 1978 | Eine Bemerkung zum Hasse-Syracuse Algorithmus |
| P | Akin | 2004 | Why is the $3x + 1$ Problem Hard? |
| G | Mignosi | 1995 | On a Generalization of the $3x + 1$ Problem |
| V | Kascak | 1992 | Small universal one-state linear operator algorithms |
| - | Jarvis | 1989 | 13, 31 and the $3x + 1$ problem |
| V | Leavens & Vermeulen | 1992 | $3x + 1$ Search Programs |
| V | Gonnet | 1991 | Computations on the $3n + 1$ Conjecture |
| G | Leigh | 1986 | A Markov process underlying the generalized Syracuse algorithm |
| - | Rhin | 1987 | Approximants de Pade et mesures effectives d'irrationalite |
| - | Rhin & Toffin | 1986 | Approximants de Pade simultanes de logarithmes |
| - | Venturini | 1982 | Sul Comportamento delle Iterazioni di Alcune Funzioni Numeriche |
| - | Venturini | 1989 | On the $3x + 1$ Problem |
| G | Venturini | 1992 | Iterates of Number Theoretic Functions with Periodic Rational Coef ... |
| G | Venturini | 1997 | On a Generalization of the $3x + 1$ problem |
| - | Hasenjager | 1990 | Hasse's Syracuse-Problem und die Rolle der Basen |
| L | Scollo | 2005 | *omega*-rewriting the Collatz problem |
| - | Gao | 1993 | On consecutive numbers of the same height in the Collatz problem |
| - | Targonski | 1991 | Open questions about KW-orbits and iterative roots |
| - | Wirsching | 1993 | An Improved Estimate Concerning $3N + 1$ Predecessor Sets |
| G | Wirsching | 1994 | A Markov chain underlying the backward Syracuse algorithm |
| - | Wirsching | 1996 | On the Combinatorial Structure of $3N + 1$ Predecessor Sets |
| P | Wirsching | 1996 | $3n + 1$ Predecessor Densities and Uniform Distribution in *mathbb* ... |
| P | Wirsching | 1998 | The Dynamical System Generated by the $3n + 1$ Function |
| - | Wirsching | 1998 | Balls in Constrained Urns and Cantor-Like Sets |
| - | Wirsching | 2000 | Uber das $3n + 1$ Problem |
| V | Glaser & Weigand | 1989 | Das-ULAM Problem-Computergestutze Entdeckungen |
| - | Riele | 1983 | Problem 669 |
| - | Riele | 1983 | Iteration of Number-theoretic functions |
| - | Coxeter | 1971 | Cyclic Sequences and Frieze Patterns, (The Fourth Felix Behrend Me ... |
| - | He & Wang | 2003 | $3n + 1$ problem's simplifying and structure property of $T(n)$ ... |
| - | Lunkenheimer | 1988 | Eine kleine Untersuchung zu einem zahlentheoretischen Problem |
| - | Muller | 1991 | Das '$3n + 1$' Problem |
| - | Muller | 1994 | Uber eine Klasse 2-adischer Funktionen im Zusammenhang mit dem " ... |
| S | Hasse | 1975 | Unsolved Problems in Elementary Number Theory |
| V | Moller | 1978 | Uber Hasses Verallgemeinerung des Syracuse-Algorithmus (Kakutanis ... |
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